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आज़ादी का
अमृत महोत्सव

10
Standard



Mathematics Around Us

Practice Book for Students

Central Board of Secondary Education

In collaboration with



Preface

Dear Teachers

The Central Board of Secondary Education, in its ongoing journey to improve the quality of education, has developed the present set of resource books of Mathematics for Classes 6-10 titled Mathematics Around Us - Practice Book for Students. These workbooks, focussing on key mathematical competencies, have been developed by CBSE in collaboration with ThinkTac.

The resolve of the National Education Policy (2020) to make education more meaningful and relevant finds expression in the Board's continuous endeavour to shift focus from rote learning to development of competencies. Development of competencies would equip student with key 21st century skills as envisioned by NEP (2020).

The practice books present a series of questions organised as themes, mapped to concepts from the curriculum of the respective class. Each theme has multiple case studies describing a real-life situation. The questions following the cases require students to recall and apply concepts learnt in the curriculum. Implicit in it is the ability to interpret mathematical data and draw inferences. As students engage with this resource material, they would not only strengthen mathematical skills and competencies but also appreciate the relevance and applications of the concepts studied in the natural contexts. The workbooks are designed to help learners develop the ability to translate a real life problem as a mathematical problem, recall and apply mathematical processes and tools, interpret the mathematical results and draw inferences. Dialogue and collaboration with peers and teachers as students work on the theme would facilitate a deeper understanding.

Most real-life situations involve multiple concepts, often across academic subjects and classes. As the workbooks are based on exploration of real-life scenarios, a few questions may be from the curriculum of a lower or higher class. Adequate background information is provided in such cases.

We welcome feedback for further improvement.

Team CBSE

About CBSE

The Central Board of Secondary Education (CBSE) is a national Board under the Ministry of Education, Govt. of India. The Board has more than 28,000 schools affiliated to it in India and overseas. These include the Kendriya Vidyalayas, the Jawahar Navodaya Vidyalayas, Central Tibetan Schools, schools run/aided by the State Governments and private independent schools. The Board's main objective is to foster a conducive environment in schools that encourages quality education and focuses on the holistic development of children. Through progressive and innovative pedagogies, relevant to the various contexts in India, the CBSE motivates schools and teachers to adopt child-centric enquiry-based techniques that allow children to achieve academic excellence.

About ThinkTac

ThinkTac is a social enterprise that integrates experiential learning and 21st century skill development into the academic curriculum to empower a learner to build skills that enables her/him to excel in life. ThinkTac offers STEM learning programmes, integrated with I – X grade curriculum and 21st century competencies offering material kits, resources and teacher empowerment to schools.

The programmes are built on a foundation of hands-on activities - built using simple, repurposed materials offering incredible scope for building, rebuilding and tinkering, which fosters creative, problem-solvers of tomorrow. This supports children to create, experiment, tinker, innovate and learn, mastering concepts and developing competencies needed to thrive in the 21st century, enabling them to choose their life purpose.

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Theme 1: Real Numbers



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Euclid's division algorithm
- The Fundamental Theorem of Arithmetic
- Finding HCF and LCM by prime factorisation method
- Rational and Irrational numbers



Water Tankers

In many cities across the country, we often get to see water tankers transporting water from one part of the city to another. Tractors are often used for doing this job. Tractors are heavy-duty vehicles custom designed to move without getting stuck in soft or damp soil. This makes them suitable for agricultural use. These tractors have now been adapted for alternate uses, like transporting water.

Case Study A - Tractor Tyres

Tractors have rear wheels, which are bigger in diameter than the front wheels. The rear wheels have diameters in the range of 45 to 55 inches (around 115 cm to 140 cm). Note that this is the outer diameter, i.e. it includes the height of the tyre.



Fig. 1.1, Tractor (Water tanker); Image by Rakesh.5suthar via Wikimedia Commons

The front wheels have outer diameters in the range of 25 to 30 inches (around 65 cm to 75 cm). This difference in the size of wheels is not commonly seen in other vehicles. In tractors, the larger rear wheel shifts the centre of gravity closer to the rear wheel, where the driver's seat is located. This gives it the stability it needs while driving on uneven surfaces, like fields or uneven roads. The short front wheel gives good visibility to the driver from the high positioned seat.

When any object comes in physical contact with another and a movement of either of the two objects occurs, the two objects undergo some form of wear and tear due to the rubbing at the surface level. More frequently a surface comes in contact with another, more is the wear and tear. The tractor tyres come in frequent contact with land, and with hard and uneven surfaces, they are more likely to experience wear and tear. The uneven surfaces exert high pressure on specific points of the tyres. The tractor tyres are extremely strong by design to minimise the rate of wear and tear.

Will the wear and tear experienced by the front tyres be the same as the rear tyres?

Question 1

- i. Consider a tractor with the rear wheel having an outer diameter of 49 inches and front wheel 28 inches. When the tractor is stationary, each of its wheels is touching the road at one point. Let this point on one out of the two front wheels be point A. Let us name the point on one of the two rear wheels as point B. As the tractor starts moving, so do the points A and B.

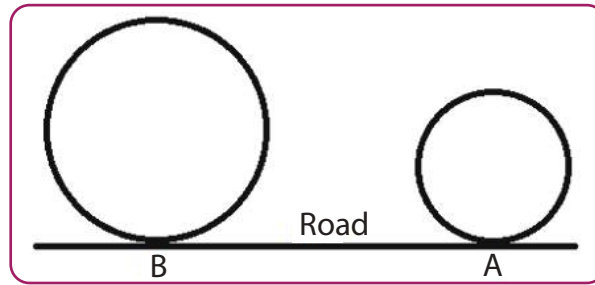


Fig. 1.2, Tractor wheels and point of contact with road

After how many rotations of the rear wheel and front wheel will both these two points touch the road again at the same time? Write your answer in the space provided below.

a. Number of rotations of rear-wheel: _____

b. Number of rotations of front-wheel: _____

ii. While solving a mathematical problem, using a method that involves the least number of steps may save time & effort. Which of the following steps can be skipped for answering the previous question?

- a. Calculate the radius of the front wheel and rear wheel
- b. Calculate the circumference of the front wheel and rear wheel
- c. Finding the LCM of two numbers
- d. Dividing one number by another number

Answer

Question 2

The number of times point A touches the road in one rotation is the same as the number of times other points on the front wheel's tyre touch the road (in one rotation). Same for point B on the rear wheel's tyre. (Use $\pi = \frac{22}{7}$).

i. After the tractor has moved a distance of 1 km, how many times would point A have touched the road? What about point B? Write your answer in the space provided below. Note that 1 km = 39,370 inches.

Answer

- ii. Which tyres are likely to undergo more wear and tear - rear tyres or front tyres? Assume that the thickness of the front and rear wheel tyres are the same, and so is the quality of rubber used to manufacture them. Write your answer in the space provided below.

Answer

Case Study B - School Auditorium

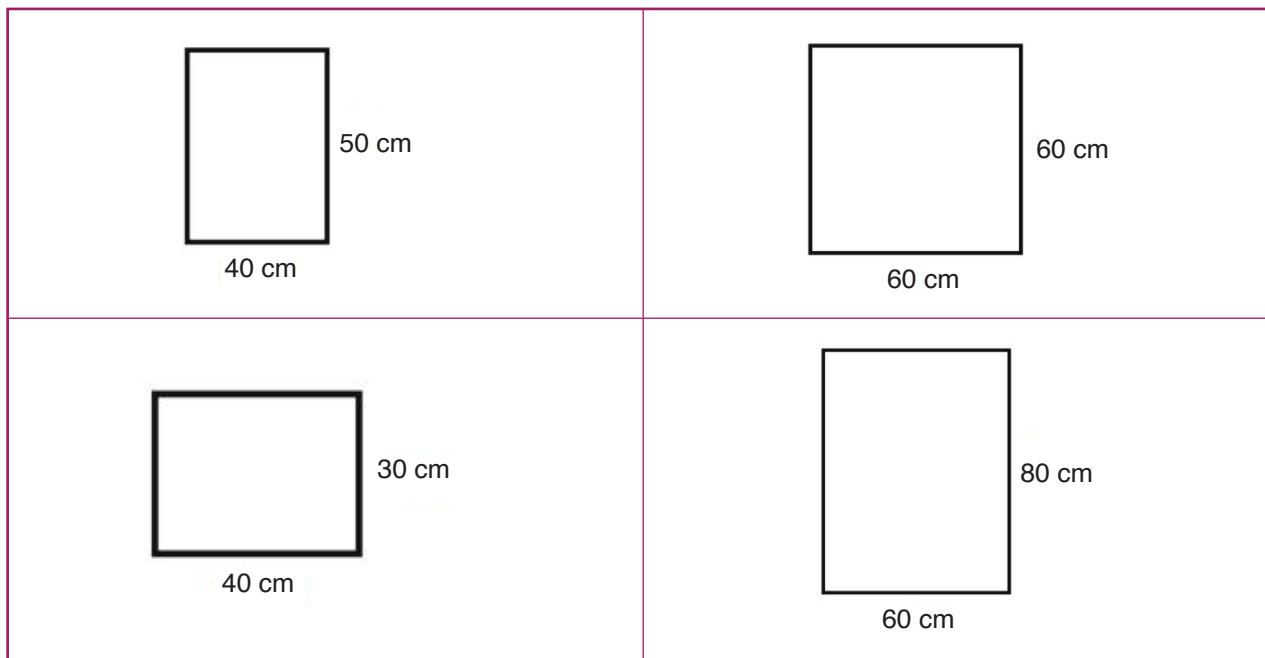
A school auditorium has a rectangular stage, which is 9.6 metres long, 15.2 metres wide and is to be paved. The tiles available are all square-shaped. A masonry team is asked to select the most suitable tiles. The team reasoned that the effort required to lay bigger tiles is less than that for laying smaller tiles. The team was informed that the material used to make these tiles has the property of splitting unevenly if one tries to break the tile into smaller pieces. So, in order to minimise wastage and also to maintain a smooth transition from the front stage to the backstage, the masonry team has to work with the constraint of selecting a size, which covers the entire stage exactly, without the need to break any tile.

Question 3

- i. The team has to select one size from the following. Which one should they select? Only one side of each size is specified (in cm), since these are square-shaped tiles.

a. 20	b. 10	Answer
c. 40	d. 50	

- ii. On researching further, the team found another vendor, who offered a few more options, which included rectangle-shaped tiles.



Do you think they should continue with the decision to use the previously chosen square tile or change it? Explain your answer in the space given below and include the new size in your explanation, if you think they should change their decision. Assume that the total cost does not change significantly if they change the type of the tile. Also, assume that the constraint of not breaking the tiles into smaller pieces continues to hold.

Answer

Question 4

The school invites parents as well as students from a neighbouring school, during the annual function. The seating of parents and students was arranged in separate rows. The teachers tried the following combinations:

- 16 students per row, they were left with 11 students
- 20 per row, left with 15
- 25 per row, left with 20
- 32 per row, left with 27

The idea was to have an equal number of students from their own school in each row so that the rest of the seats from each row can be allocated to the students from the neighbouring school.

If you include 5 students from the neighbouring school while looking for the seating arrangement, how many students will be left for each combination? Write your answer in the blank spaces provided in Table 1.1.

Number of students per row	Number of remaining students
16	_____
20	_____
25	_____
32	_____

Table 1.1, Number of students per row and number of students remaining

Question 5

i. Based on the information available, if you want to guess the total number of students in this school, what process would you follow?

- Find the highest factor of 32, which is also a factor of 16, 20 and 25
- Find the first common multiple of 16, 20, 25 and 32 to arrive at the answer
- Find the first multiple of 16, which is divisible by 20, 25 and 32; then subtract 5 from this value to arrive at the final answer
- Find the first multiple of 16, which is divisible by 20, 25 and 32; then add 5 to this value to arrive at the final answer

Answer

ii. If the team who is looking after seating arrangement wants to continue to work with the rule of the same number of students from their school seated in each row, what is the largest number which will satisfy this constraint? Each row in the auditorium has a maximum of 60 seats. Write your answer in the space provided below.

Answer

Case Study C - Practice Schedule

Many countries have practice grounds where football and cricket are played on the same site. Consider one such ground where cricket and football are practised, but the cricket team has more practices scheduled, due to an upcoming match. These two teams start their practices on March 9th. The cricket team meets for practice every 2 days, and the football team meets every 4 days.

Question 6

For how many days will the teams not meet each other before they share the field again, and how many times will they have to share the field till the end of March, including Sundays and excluding March 9th?

a. 3, 6	b. 2, 6	Answer
c. 3, 5	d. 2, 7	

Question 7

The organising committee decided to distribute free goodies on the occasion of 100th match being played on the ground for online tickets only. Everyone who purchases the ticket will receive a bag with a small gift. Online booking programs are written in such a way that everyone does not receive a referral code to claim the gift. Table 1.2 shows which ticket numbers will receive a gift.

Items	Ticket number
Pen	Every 2 nd ticket
Cap	Every 7 th ticket
Key chain	Every 10 th ticket

Table 1.2, Gift items as per the ticket number

A family book their tickets online and they are allotted seat numbers - 74, 75, 76, 77. Would anyone from this family get all the three gift items? If yes, which seat number is the lucky one? If not, how many types of items will they get in total as a family?

- a. No, they get only two types of items in total
- b. No, they get only one type of item in total
- c. Yes, seat number 77
- d. Overall they will have all 3 types of items but not through a single ticket

Answer

Case Study D - Patterns

Here is an interesting activity, which can lead to beautiful pattern(s).

1. Take a sheet and draw the x and y axis such that the origin is at the centre of the page (approximately).
2. Draw an arc intersecting the x-axis with the origin as the centre, in the positive direction. The radius of the arc should be 2 cm.
3. Now draw a line perpendicular to the x - axis at the point where the arc meets the x-axis. On this perpendicular line draw another arc which has a radius 2 cm.
4. Join the point where the arc intersects the perpendicular line to the origin. This line ($\overline{OC}$), the perpendicular line ($\overline{BC}$) and the line on the x-axis ($\overline{OB}$) form a right angled triangle $\triangle OBC$, as shown in Fig. 1.3.

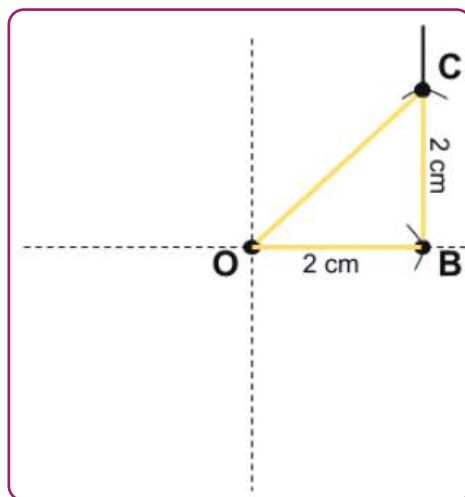


Fig. 1.3, Triangle constructed by arcs of equal length

5. Now draw a line perpendicular to $\overline{OC}$ at C and draw an arc 2 cm long on this line with C as the centre, as shown in Fig. 1.4. This will form another right angled triangle $\triangle OCD$.

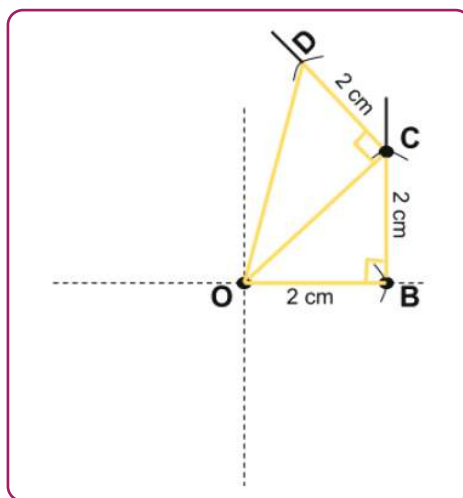


Fig. 1.4, Triangles constructed by arcs of equal length

6. Follow this rule and draw another 23 triangles to form a pattern with 25 triangles. The next triangle in this series will have the line $\overline{OD}$ as its base and a 2 cm long line perpendicular to $\overline{OD}$.

7. We will refer to the $\triangle OBC$ as Triangle 1, $\triangle OCD$ as Triangle 2 and so on.

A larger pattern with several triangles is given in Fig. 1.5

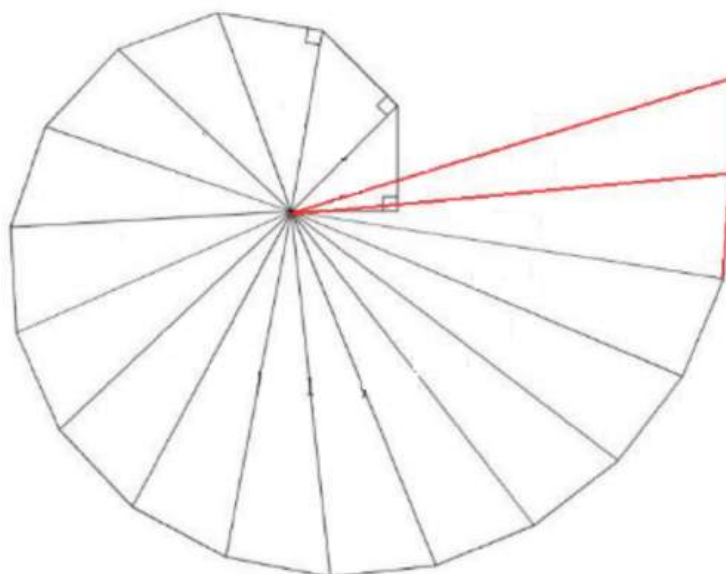


Fig. 1.5, A larger pattern with many triangles

Question 8

In $\triangle OBC$ we shall call the line $\overline{OB}$ as the base and line $\overline{BC}$ as the perpendicular. Let us follow the same naming convention for all the 25 triangles. The length of the base and perpendicular of Triangle 1 ($\triangle OBC$), is a rational number. What about the base and perpendicular of Triangle 2 and all the remaining triangles?

Fill Table 1.3 with your prediction about the hypotenuse, base and perpendicular of all the 25 triangles. Eg: If you think that the hypotenuse of the triangles will be a rational number for all the 25 triangles, then write “Y” under “Always Rational”. If you think it will be a rational number for some triangles but irrational for others, write “Y” under “Sometimes Rational, Sometimes Irrational”.

	Always Rational	Always Irrational	Sometimes Rational, Sometimes Irrational
Hypotenuse			
Perpendicular			
Base			

Table 1.3, Prediction about the type of sides of the triangles

Question 9

As we add more triangles to this pattern, will the pattern grow or shrink? To understand this, let us check if the sides and angles of the triangles increase as we add more triangles or decrease.

For each triangle in the pattern discussed above, let us call the angle formed by the base and hypotenuse as θ . In $\triangle OBC$ the θ will be $\angle COB$. As you go from one triangle to the next one in the series of triangles, what happens to the length of the sides of the triangle - increases/ decreases/ remains constant? What happens to θ ? Write the answers to these questions in the blank space next to each side and angle.

- Base _____ (increases/decreases/remains constant)
- Hypotenuse _____ (increases/ decreases/remains constant)
- Perpendicular _____ (increases/decreases/remains constant)
- θ _____ (increases/decreases/remains constant)

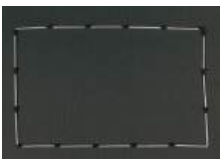
Question 10

Analyse the following statements and mark them as always true, never true and some times true.

Answer

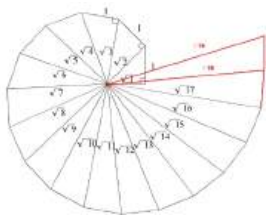
- If the radius of a circle is rational, then the area must be irrational.
- If the area of a circle is rational, then the circumference is rational.
- If the circumference of a circle is rational, then the area is rational.
- The diagonal of a square is irrational.
- The hypotenuse of an isosceles right triangle with a whole number as its base, is always irrational

Exploration Pathway



Area, Perimeter
& Factorisation

Pick any number smaller than 50 and make rectangles using matchsticks such that the area of the rectangle is equal to the number you have selected. Repeat this exercise for 5 numbers and record the length of sides of the rectangles you made. Identify the numbers for which you can't make more than one size of rectangles.



Square Root Spiral

On a graph paper, draw the x and y axis. From the origin, draw an arc of length 2 cm on the x axis. Draw a line from the point where the arc intersects the x axis. The line should be perpendicular to the x axis. Now draw another arc on this perpendicular line, of length 2 cm. Join the point of intersection of this arc to the origin. This would form a right angled triangle with two sides 2 cm long and the hypotenuse $2\sqrt{2}$ cm long.

Now draw the next right angled triangle by following the same steps and using this hypotenuse as the base of the next triangle. Continue these steps and draw 20 triangles. Observe and record the length of the base, perpendicular and hypotenuse of any 5 triangles. Do these three lengths follow a pattern? Are they always rational/irrational or only sometimes rational/irrational? Which of these lengths are growing/shrinking as you go to the next triangle?



Irrational Numbers -
Paper Sizes

When we fold an A4 standard sized paper into half along its longer length, we get the A5 standard size. This pattern continues as you go from A5 to A6 or from A0 to A1. This pattern in the standards emerges from a requirement to enlarge/reduce the size of an image at the printer.

In this TACTivity we use a bunch of A4 papers to visualise all the standards in the A series of papers starting from A0 to A8. We also observe the ratio of the sides of each standard and identify the pattern.

Theme 2: Arithmetic progressions



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Term, common difference
- General form of an AP
- n^{th} term of an AP
- Sum of first n terms of an AP



Amphitheatre

Most of us have seen an amphitheatre in some context or another: either at school, or at a performance venue, or even temporary ones at fairs and *melas*. The name itself has Greek and Roman origins, and famous amphitheatres like the Coliseum in Rome or Epidaurus in Greece come to mind.

They tend to be of a similar shape, inspiring many modern theatres too: a curved, even semicircular, arrangement of seats, tiered at different heights, giving a clear view of a stage placed opposite the seating at a slight height/plinth; and typically set in the outdoors. This shape and arrangement gives a clear view to all spectators and also enhances the acoustics of the space. This makes an amphitheatre an ideal space for music, dance and theatre performances.

It is crucial to know the total number of seats planned in each row, to decide the entrance, exit, where to leave gaps for easy access to seats and so on, during the initial design phase of building an amphitheatre.



Fig. 2.1, Model of amphitheatre

Case Study A - Amphitheatre

Each amphitheatre has two main sections - one where the artists perform (the stage) and one where the audience is seated. While designing an amphitheatre, the architect evaluated different possible shapes for the audience section. The goal was to design an amphitheatre with 26 seats in the first row and a total of 20 rows.

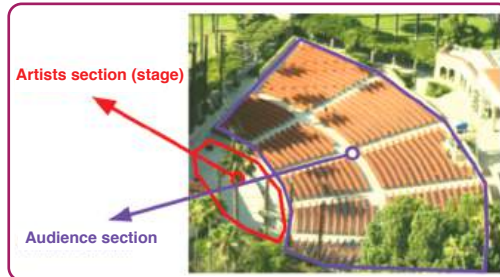
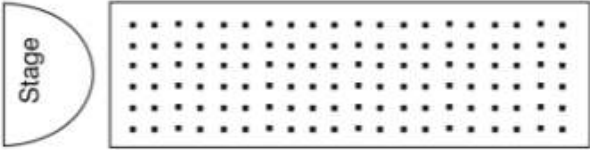
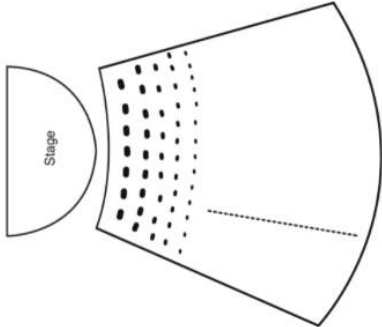
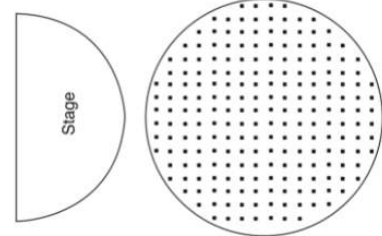


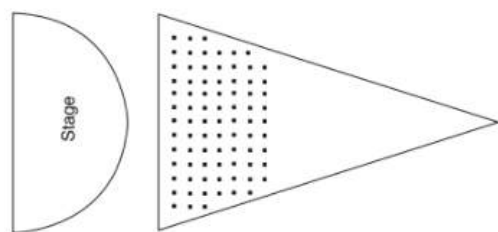
Fig. 2.2, Picture of amphitheatre showing audience and artist sections

Question 1

Given below is a list of shapes (top view) for the audience section. For each, select the relevant formula representing the number of seats in the R^{th} row, where row numbers start with 0 and end with 19. Assume that seats are placed from one end of the wall to another in each row.

Shape for audience section	Formula
a.  Rectangle shaped audience section	A. $\{26 - 2R\}$
b.  A sector of a circle shaped audience section	B. if $(R < 10)$ then $\{26 + 2R\}$ else $\{44 - 2(R-9)\}$
c.  Circle shaped audience section	C. $\{26\}$

d.



Triangle shaped audience section with pointed end away from the stage

$$D. \{26 + 2R\}$$

Write the formula option (one out of A, B, C, D) next to each shape in the blank space given next to the shape:

i. Shape a, rectangle $\longrightarrow$ Formula _____

ii. Shape b, sector of a circle $\longrightarrow$ Formula _____

iii. Shape c, circle $\longrightarrow$ Formula _____

vi. Shape d, triangle $\longrightarrow$ Formula _____

After looking at the various options, the architect chose a shape shown in Fig. 2.3 as the shape of the audience section with 26 seats in the first row and 2 seats added in each of the subsequent rows. Answer questions 2 to 5 based on this design of the amphitheatre.

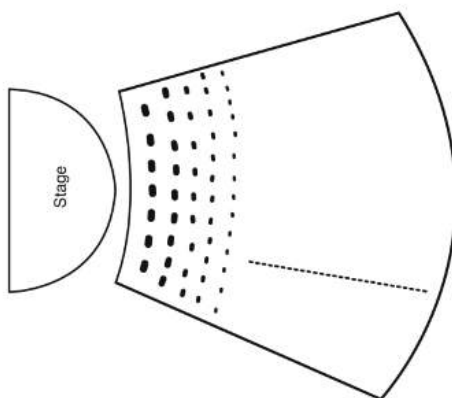


Fig. 2.3, Shape chosen by the architect

Question 2

What is the total seating capacity of the theatre?

Answer

Question 3

If one plotted a graph with the total seating capacity (till a given row number) on the y-axis and row number on the x-axis, what shape would the graph have?

- a. straight line sloping upwards
- b. flat line with zero slope
- c. parabola
- d. straight line sloping downwards

Answer

Question 4

The owners wanted to double the capacity of the amphitheatre. What is the minimum number of additional rows they need to add?

- a. 20
- b. more than 20
- c. less than 20 but more than 5
- d. less than or equal to 5

Answer

Question 5

Being open to the sky, the access to the theatre was free for birds. Bird droppings became a common phenomenon and with that came a few seeds. After a few years, 4 trees grew in the middle of the amphitheatre. The owners wanted to maintain the natural look and chose to remove the seats instead of trees. Each tree occupied the space for 2 seats per row and the trunk spread across 2 rows. (Refer Fig. 2.4)

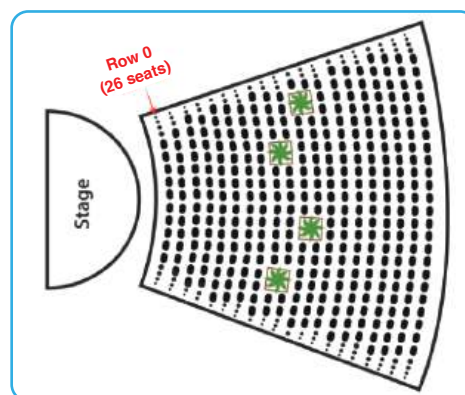
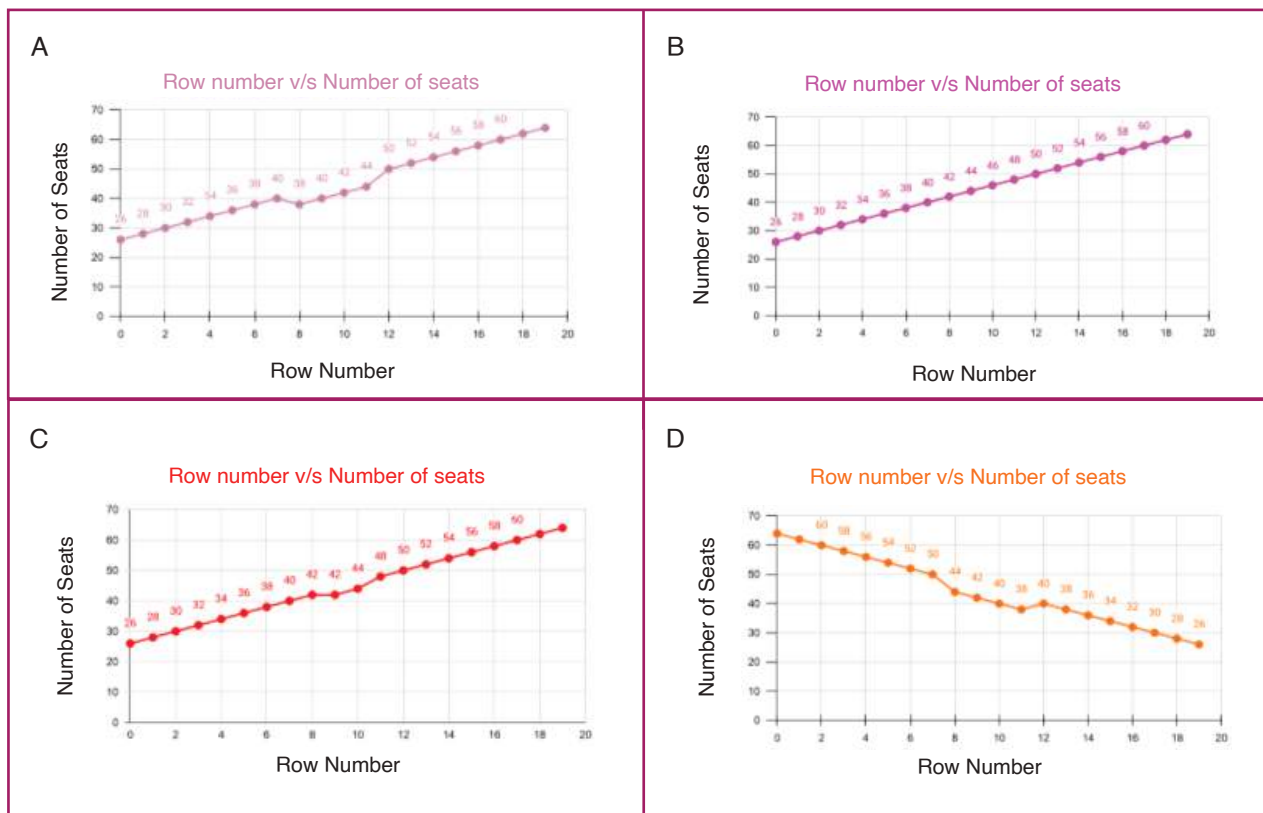


Fig. 2.4, Trees in the middle of the amphitheatre

Given below are graphs with the number of seats in each row on the y-axis and row number on the x-axis. Which of the 4 graphs (A to D) correctly represents the amphitheatre being discussed in Fig. 2.4?



Answer

Case Study B - The Pattern Game

In a funfair, there is a counter with a 1 minute game that requires participants to create patterns with sticks. Two players participate in each batch of the game. Both the players will start at same time to make the patterns similar to the one shown in Fig. 2.12.

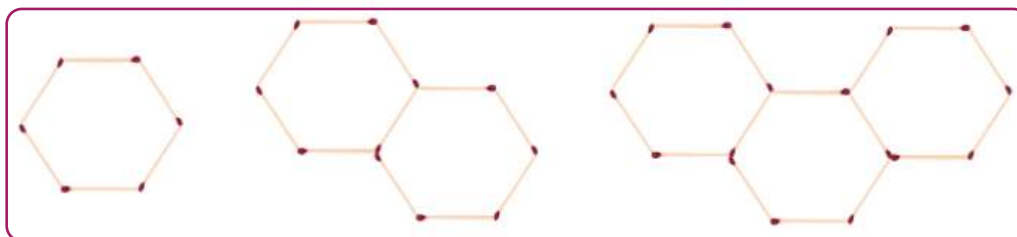


Fig. 2.5, Patterns from stick

Each participant can use any number of sticks from the tray. There should be only one common stick between the two connected patterns and each pattern should connect to the adjacent one only, making it a chain. The one who makes the longest complete pattern chain in 1 minute wins the game, where the length of the pattern chain is defined as below:

Chain length = Number of complete hexagons in the chain

Question 6

Aditi and Renu participate in this game. Aditi uses 61 sticks and Renu makes a chain with 12 hexagons in one minute. Who gets the prize?

- a. Aditi, because she uses more sticks
- b. Renu, because she makes more patterns
- c. Aditi, because she gets chain length as 13 by using 61 sticks
- d. Both share the prize equally, because the chain length is the same

Answer

Question 7

- i. If adding each hexagonal pattern, including the first one, takes a minimum of 4 seconds, what is the maximum number of sticks that can be used by a player in this one minute game? Write your answer in the space provided.

Answer

- ii. How will the number calculated above help the organisers of the funfair counter?

- a. Estimate the maximum number of batches they can run in the 4 hours available
- b. Estimate number of prizes they need to arrange
- c. It is not helpful, since different players will take different time to make the patterns
- d. Decide how many sticks to place in the tray

Answer

Question 8

As more players play the game, the word starts spreading and people start queuing at the counter. The organisers have 10 people in their team, who can manage the game, and hence they decide to run 5 games in parallel batches. But they have a limited stock of sticks. What is the minimum number of sticks they need to run 5 batches in parallel?

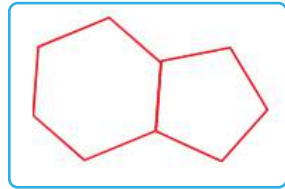
- a. Upto 500
- b. More than 500 but less than 750
- c. 750 to 900
- d. More than 900 but less than 1100

Answer

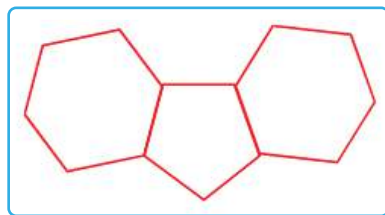
Question 9

To give a twist, the fun fair management team made a small change after a few rounds of the competition. Every alternate shape added through the sticks will be a pentagon. This means

- a chain of 2 will have 1 hexagon + 1 pentagon

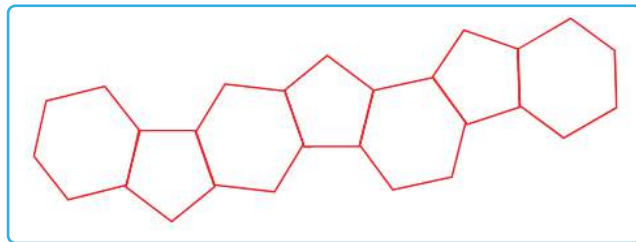


- chain of 3 will have 1 hexagon + 1 pentagon + 1 hexagon



Continuing in the same manner,

- chain of 7 will have 1 hexagon + 1 pentagon + 1 hexagon + 1 pentagon + 1 hexagon + 1 pentagon + 1 hexagon



Write a formula in the space provided below, representing the total number of sticks required for a chain of length P , where P is an:

i. odd number

Answer

ii. even number

Answer

Case Study C - Regular Polygons

Our life is surrounded by a variety of shapes. The 2D shapes, which are made of three or more straight lines joined together from end to end to enclose a space, are called polygons. The polygons, which have all their sides of equal length and all angles of equal measure, are called regular polygons. Regular polygons often look pleasing to the eyes due to their symmetry. As we go from an equilateral triangle to a square to a regular pentagon (and so on), we observe patterns in the size of angles, sum of angles, etc. Let us explore which of these patterns have an Arithmetic Progression (AP).

An interior angle of a polygon is the angle formed inside the polygon by any two adjacent sides of the polygon. An exterior angle is the angle formed outside the polygon by one side and extending one of its adjacent sides. The sum of the interior angle and the exterior angle on the same vertex is always 180° .

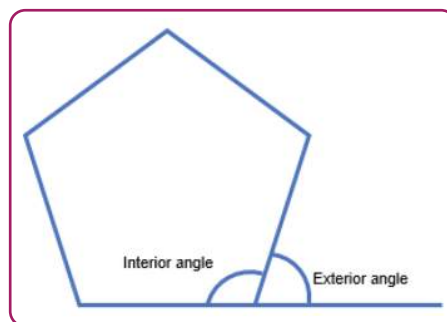


Fig. 2.6, Interior and exterior angle example in a pentagon

How to calculate the interior angle of a regular polygon?

1. If you don't know the value of an interior angle of a regular polygon, you can calculate it by considering the fact that the interior angle is the supplement of the exterior angle.
2. Recall that the sum of exterior angles of a regular polygon is 360° .
3. Each interior angle = $180^\circ - \left(\frac{360^\circ}{n}\right)$, where n is the number of sides of the polygon, $n \geq 3$.

Question 10

Complete table 2.1. Fill column 7, 8 and 9 only if the answer to the 6th column is "Yes" for a particular row else fill NA (not applicable).

Column #1	#2	#3	#4	#5	#6	#7	#8	#9
Regular polygon	Equilateral triangle	Square	Regular pentagon	Regular hexagon	Is this row in AP? (Yes/No)	First term (a)	Common difference (d)	n th term
	Term 1 (a ₁)	Term 2 (a ₂)	Term 3 (a ₃)	Term 4 (a ₄)				
No. of sides	3	4	5	6	Yes	3	1	n+2
Interior angle	60							
Exterior angle	120							
Σ Interior angles	180							
Σ Exterior angles	360							

Table 2.1, Table of angles and sides of regular polygons

Exploration Pathway



Growing Patterns

Make a hexagon, using 6 matchsticks. Add another 5 sticks to attach another hexagon, which shares one side with the first hexagon. Repeat this process and make different patterns. Can you make a pattern with more than 2 hexagons, where the last hexagon shares a side with the first hexagon? Predict and validate the number of sticks required to make 15 hexagons.



DIY Buckminsterfullerene Model

Take a few stiff straws and join them together, using loops of binding wire. Prepare hexagons and pentagons and join them together to make your own Buckminsterfullerene model. Observe the pattern in the design and predict the number of wire loops you will need to join the straws. Also predict the number of straws required to make the full model.

Theme 3: Triangles and Similarity



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Basics of Geometry
- Types of Triangles
- Lines and Angles
- Triangle Theorems



Scientific Modelling

Other than microscopes, telescopes and projectors, how often and where do we see magnification in real life? Imagine an Engineer is constructing a house for one of her clients. She wants to take her client's inputs and decide on the size of the rooms, the shape of the building, etc. Now imagine she constructs some sample rooms and a sample building so that her client can make a selection. Does that sound absurd? Be it constructing a new building or designing a new product or the replacement of a body part or analysing the route from one point to another, modelling the actual product helps in planning, analysing and evaluating before making the product or instead of physically visiting the route. One type of model that we work with are called maps. You may draw the map of your hand by placing your palm on a piece of paper. This map can be carried around by someone to say buy hand gloves, which fit your hand, without taking you or your hand around. This would be a map, which is as big as your hand but you may as well draw maps which are bigger/smaller than the actual object/area they are mapping, like the example of a building or a route map. In such cases, we use a scale to magnify or minify and still represent the physical object/area and communicate the information effectively.

Case Study A - Scale Factor

A company acquires 6 hectares of agricultural land for cultivating 5 types of cash crops. The shape of the land is a perfect rectangle with one side being 300 m long. To plan the distribution of the 5 types of crops on this land, a map has to be prepared. Imagine that the plan is prepared on an A4 paper.

Note:

1. Consider the size of an A4 paper as 30 cm x 21 cm.
2. The goal is to utilise as much part of the A4 paper as possible so as to get the maximum clarity in the map.
3. 1 Hectare is an area occupied by a square with the side equal to 100 m.
4. A scale for a map is specified as a ratio, something like 1:10. The first number here (1 in this case) indicates the length in the map and the second (10 in this case) indicates the length in the physical world. For this example, a line/curve which is 1 cm long on the map will be 10 cm long in the physical world.

Question 1

- i. Which of the following scales should be used? Select the best option out of those given below:

a. 1:120	b. 1:10	Answer
c. 1:1200	d. 10:1	

- ii. What is the ratio of the area of the actual land bought by the company to the area of its map depicted on an A4 sheet? Assume that the map of the land is drawn in such a way that the maximum possible area of the A4 sheet is utilized to draw the map.

a. 1:100	b. 1:1000	Answer
c. 1:10000	d. 1:1000000	

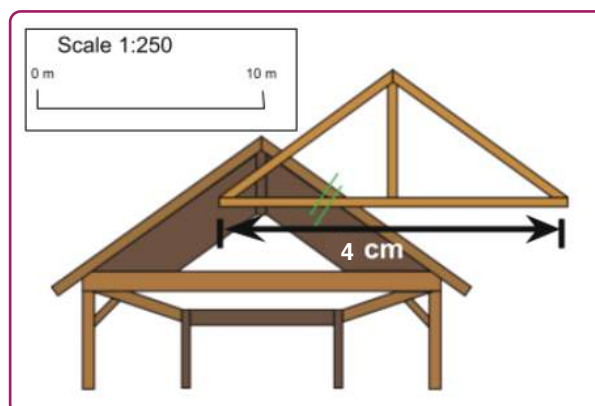
Question 2

Fig. 3.1, House with sloping roof

Fig. 3.1 shows a model of a house, which has a sloping roof. It has a triangular shaped frame below the slope. Find the actual width of the roof, in metres. Write your answer in the space provided.

Answer

Question 3

The triangular section of the wall right under a roof is called a 'gable' as shown in Fig.3.2.

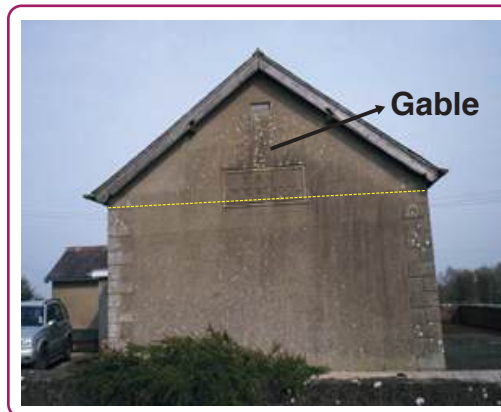


Fig. 3.2, A building with a gable; Image by P Flannagan via Wikimedia Commons

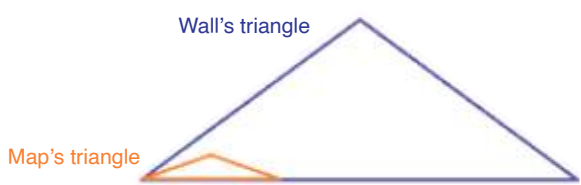
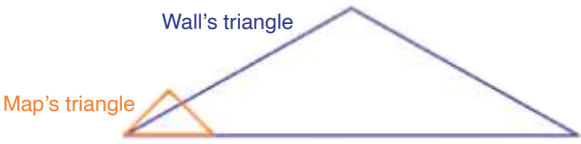
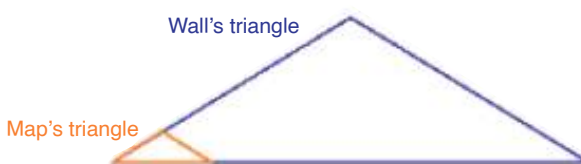
i. If you compare the gable of the actual house mentioned in Question 2, with its model in the map, how do you think the two triangles are related?

- a. These two triangles are congruent
- b. They both are similar
- c. Neither similar nor congruent

Answer

ii. Predict what you will see if you cut out the triangle from the map and stick it on any one corner of the actual triangular wall. Refer to the picture next to each option (Fig. 3.3-3.5) to visualise each option.

Note that we do not rotate the map's triangle before sticking, which means the apex of the triangle would always be on the top. In our case, the apex is the vertex of the triangle representing the corner where the two roofs connect to each other.

a. The map's triangle will not fully cover the corner of the wall where it is being pasted.	 Fig. 3.3, Answer option a
b. Some part of the map's triangle will go outside the wall.	 Fig. 3.4, Answer option b
c. Map's triangle will exactly overlap with the wall's corner.	 Fig. 3.5, Answer option c

Answer



Pinhole Camera

A pinhole camera is a simple device where light enters a box through a small aperture and an image is projected on a screen without the need for a lens. You can easily make a pinhole camera using a chart paper, butter paper and tape to see lovely inverted images. This camera doesn't have a lens but just a tiny aperture. When light passes through this tiny aperture, it projects an image of the object being observed through the aperture. You can change the size of the image by varying the relative distance between the aperture (pinhole) and the screen.

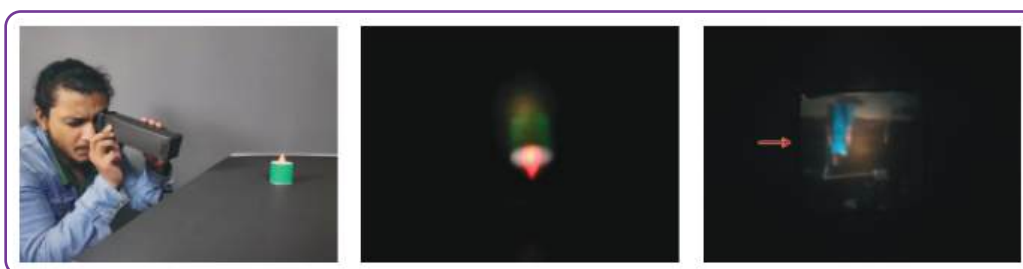


Fig.3.6, Using the pinhole camera and sample images as seen through the camera

Implement a pinhole camera yourself and experience the change in the size of the image (if any) as you move the outer box of the camera. To access a quick guide for this activity please visit ThinkTac's website or YouTube channel.

Case Study B - Light Through Small Apertures

One of the standard ways of drawing diagrams to understand the behaviour of light is to draw rays coming from the topmost point of an object and those coming from the bottom.

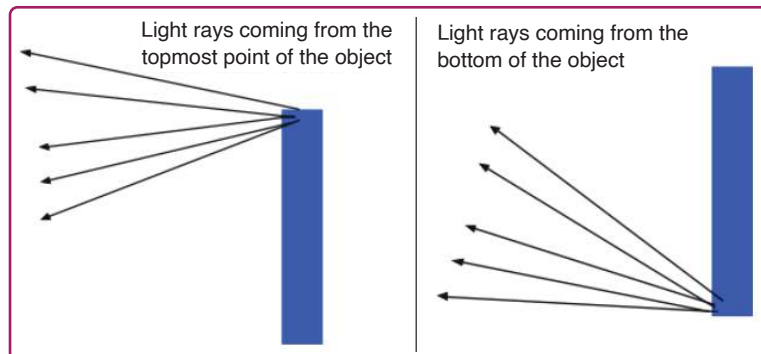


Fig.3.7, Rays of light coming from an object represented as straight lines

Now, imagine placing an opaque sheet in front of the object, which will stop light from passing through and then make a hole in the sheet which is so small that only one ray of light can pass through.

Here, if we include the object and image, then we see two triangles on the two sides of the opaque sheet. Let us look at this type of camera through the lens of the triangles formed with the light rays.

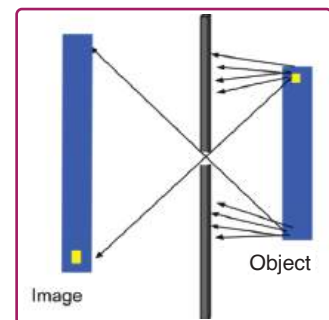


Fig.3.8, Rays of light passing through a hole in an opaque sheet forming two triangles

Question 4

In the pinhole camera diagram (Fig. 3.9), H is the actual size of the object, D is the distance from the pinhole camera aperture to the object, h is the size of the object's image in the pinhole camera, and d is the distance from the pinhole camera aperture to the image projection surface.

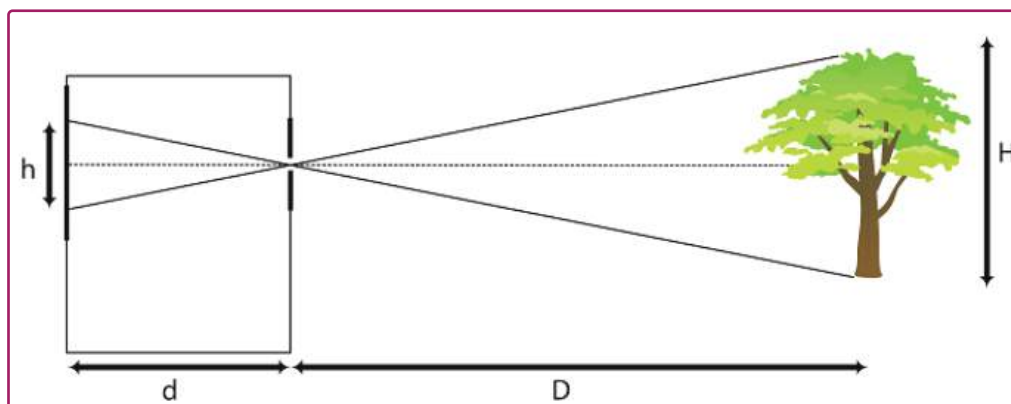


Fig.3.9, Pinhole camera schematic

Which of the following equations correctly describes the relationship between image height h , projection distance d , object distance D , and object height H ?

a. $H \times h = d \times D$	b. $H \times d = h \times D$	Answer
c. $H \times D = h \times d$	d. $\frac{h}{D} = \frac{H}{d}$	

Question 5

Suppose the distance, d , between the screen and the pinhole is doubled from 15 cm to 30 cm. How will it affect the image of an object kept at a distance of 20 cm in front of the pinhole?

- a. Image size doubles
- b. Image size reduces by a factor of 2
- c. No change of the image size as long as the object is the same
- d. Image size will be two-third of the original size

Answer

Question 6

Magnification tells us if the image is of the same size, smaller or larger than the object. It is represented by the unitless quantity, m , and it is defined as the ratio of the image height to the object height. If the image-to-object distance is double of the pinhole-to-object distance, then m will be

a. Less than 1	b. More than 1	Answer
c. Equal to 1	d. Depends on the hole size	

Case Study C - Robotics Prelims

Four grade 10 students from Pragati school participated in the preliminary round of a robotics competition. The model diagram (Fig. 3.10) was shared with all participants a week before the event, which included the details of the paths and angles in the path of the robot's movement.

The robot placed at point A is tasked to pick the can from anywhere in its path from A to C and put it in the bin located at B. Each of the four students came up with different models, using different design components. In the registration form, each one of them updated the battery life and speed at which their robot can move. Table 3.1 shows these details updated by the students.

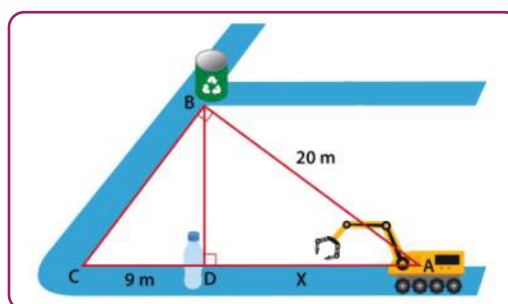


Fig. 3.10, Route map for the robot

Student name	Battery life (hours)	Robot's speed (m/s)
Sanjay	5	0.5
Renu	3	1.0
Joseph	4	0.5
Syed	6	0.2

Table 3.1, Battery life and speed of robot designed by the four students

Question 7

What is the maximum and the minimum distance the robot has to cover from its starting position A to pick the bottle anywhere in the path A to C and drop it at the bin positioned at B? Write your answer in the space provided below. Note: The bottle cannot be placed at A and the robot cannot travel backward.

Answer

Question 8

Robots are kept switched ON all the time from the start of the competition at 10 a.m. For each participant, the competition starts with a quick interview with the judges. It is followed by a demo, where the robot is expected to start from point A and come back after finishing the task in less than 3 minutes. After this, to test the sensor design, the robots are made to go around the path multiple times by slightly altering the position of the can, without changing the position of the bin. With multiple schools participating in the competition, Pragati School students get a chance to demonstrate only at 12:30 p.m. Who all (if any) will get disqualified because of their robot not coming back within the 3 minute time limit? Write your answer in the space provided. If nobody misses the chance, write "None". Justify your answer.

Answer

Case Study D - Three Triangle Logo

The logo of a company comprised of a series of 3 similar triangles. Fig. 3.11 shows the 3 triangles. Note that this figure has not been drawn to scale.

All the angles at A are equal. Also, the ratio of some sides of the triangles are equal:

$$\frac{AB}{AG} = \frac{AG}{AF} = \frac{AF}{AH} = \frac{3}{5}$$

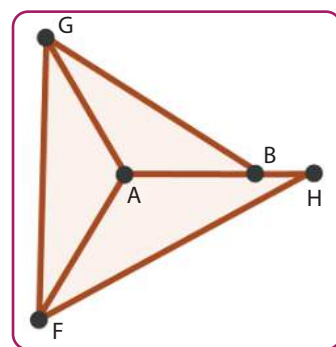


Fig. 3.11, Spiral logo of the company with 3 similar triangles

Question 9

Unlike what is shown in Fig. 3.11, the company wanted to redesign their logo to showcase their 70% growth. The company's graphic designer provided a constraint that the logo must fit within a circle of radius 7 cm. Do you think this constraint can be satisfied if the design team chooses to have the length of the side $\overline{AB}$ of the triangles as 2.7 cm?

- Yes, the constraint of fitting the logo within a circle of radius 7 cm can be satisfied
- Yes, actually the logo can be fit inside the circle of radius 6 cm
- No, it is not possible to fit the logo inside the circle of radius 7 cm
- No, it can't be fit inside of the circle of radius 7 cm, but it can be fit inside of the circle of radius 8 cm

Answer

Question 10

Given that the logo has three similar triangles, each will have an angle equal to another angle in the other triangle. Table 3.2 has a list of the 3 angles in $\triangle ABG$ (in Fig. 3.11). Identify two more triangles, which are similar to $\triangle ABG$, and write the name of the corresponding angles, which are equal to the respective angle in $\triangle ABG$.

Triangle 1	Triangle 2	Triangle 3
$\triangle ABG$		
$\angle ABG$		
$\angle AGB$		
$\angle BAG$		

Table 3.2, Table placing the equal angles of the 3 similar triangles in a single row

Exploration Pathway



Pinhole camera

A pinhole camera is a simple camera without a lens but with a tiny aperture. When light passes through this tiny aperture, it projects an upside down image of the object being observed through the aperture. In this TACTivity, we will make a pinhole camera using a chart paper, butter paper, and tape to see lovely inverted images.



Flexagon

1) Make a 3D flexagon from a rectangle with sides in the ratio of 1:2. Observe the sizes of the 6 triangles in the final flexagon. Are they equilateral/isosceles/scalene?

2) Change the ratio of the rectangle to 1:4 and predict the type & angles of the 6 triangles. Make and validate your prediction.

Theme 4: Coordinate Geometry



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Interpreting a point on a given axis, distance of a point from the given axis, and collinear points
- Distance Formula
- Section Formula
- Area of triangle using co-ordinates of its vertices



Sky Ways

Consider two planes A and B travelling on different flight paths, which possibly intersect. Are they in danger of colliding midair? How would one know? The important parameters to be considered with air traffic are location, distance, speed, and separation distances with other planes or common points in the path. The location in space is described using 3 co-ordinates (x, y, z) , and co-ordinate geometry helps us to understand these parameters.

In the vicinity of airports — especially large airports, where at peak traffic as many as three aeroplane landing or takeoff operations may occur every minute — the control of aircraft in the air is a challenging but extremely important operation. The graph in Fig. 4.1 shows the leading airports worldwide in 2019, based on aircraft movements.

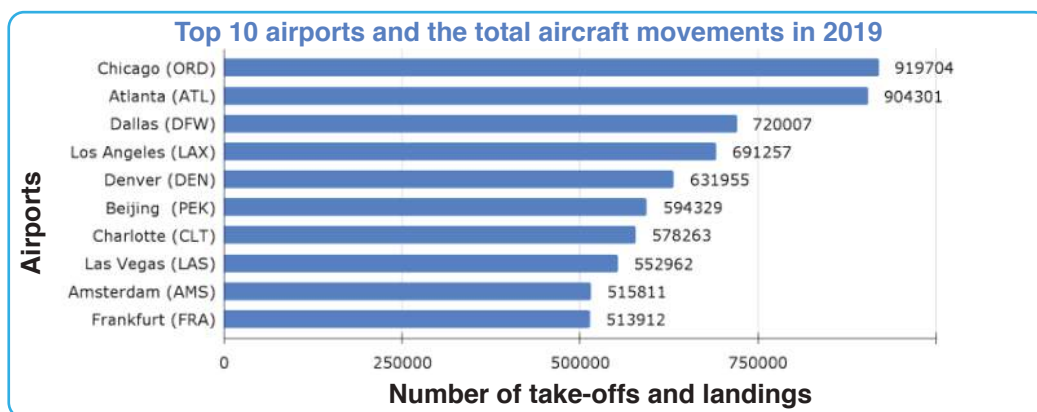


Fig. 4.1, Leading airports worldwide in 2019 based on aircraft movements data via <https://www.internationalairportreview.com/>

The altitude at which planes fly is used to make sure that planes are separated vertically too. Planes are expected to fly within a certain altitude range due to several possible factors, e.g. aircraft type, distance to the destination, weight of the airplane, type of engine, wind speeds, etc. Co-ordinating several arrivals, departures in an airport and so many flights in the sky without collisions requires masterful planning; sometimes the x and y co-ordinate seem to be close by, but the third co-ordinate, z, which is the height, keeps the flight at safe separation.

Case Study A - Air Traffic Controller

All the air traffic is controlled by the air traffic controller (ATC). ATC co-ordinates the movement of air traffic to make certain that planes stay a safe distance apart. Co-ordinate systems allow them to keep track of an aircraft's position and orientation in space. A controller system must know the location of every aircraft at any instant of time in the sky. Even if an aircraft moves a small distance (up, down, forward), its co-ordinates are updated in the system. The ATC monitors the location of aircraft by radar and communicates with pilots using radio. It's the ATC's job to ensure that planes have their own minimum airspace and that separation rules are followed between aeroplanes to prevent collisions. Consider one such air traffic controller analysing the positions of flights in an airport, using the co-ordinate system, and answer the questions below.

Question 1

Assume that each unit is 1 km to answer the following questions. The co-ordinates given below are given from a base station of XYZ airport. Find the distance between two aeroplanes, in kilometres, which are at location A and B, if A lies on the y-axis and has an ordinate 1.5, while B lies on the x-axis and has an abscissa 0.5

a. $\sqrt{2.5}$ km	b. 2 km	Answer
c. $\sqrt{2}$ km	d. 1 km	

Question 2

Three parking points are marked with co-ordinates $(2a, 4a)$, $(2a+\sqrt{3}a, 5a)$ and $(2a, 6a)$, such that these three points form the vertices of an equilateral triangle. When three planes are at these locations, what will be the distance between any two planes?

a. $4a^2$	b. $\sqrt{(2a)}$	Answer
c. $2a$	d. $2a+\sqrt{3}$	

Question 3

As you have understood, location of a plane, direction in which it is moving and speed are all important to know the possible meeting at a common point on their route. Look at the table given below (table 4.1) in which 5 scenarios are sketched out for 2 planes that are flying at the same altitude (see column IV). In each of the case the speeds of these two planes remains the same or varies (see II); in each of the case, the distance of the planes from the intersection of the routes also remains the same or varies.

The table is followed by four statements a to d. Analyse the cases and select the incorrect statement.

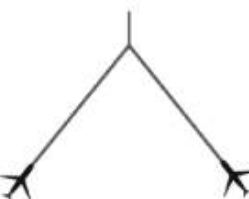
Possibility #	Plane Speeds	Distance from Intersection of Routes	Pictorial Representation
A	Different	Different	
B	Different	Same	
C	Same	Different	
D	Same	Same	
E	Same route and planes at different distance		

Table 4.1

- a. Planes may meet at the intersection in Case A and E depending on the speed
- b. Planes will meet for sure in Case D
- c. Planes will never meet in Case B and C
- d. Planes will meet for sure in Case B but not in Case D

Answer

Case Study B - Treasure Map

Disha loves reading story books. She has started reading a new series of books, which is about a pirate searching for a hidden treasure chest. She has finished reading part 2 of the book in which the location of the key of the treasure chest has been hidden in the map. Part 3 of the book will be delivered in a day or two, which will have the location given. But Disha is too impatient to wait and decides to find out the location of the key by herself. Can you help Disha find the location of the key by answering the questions below?

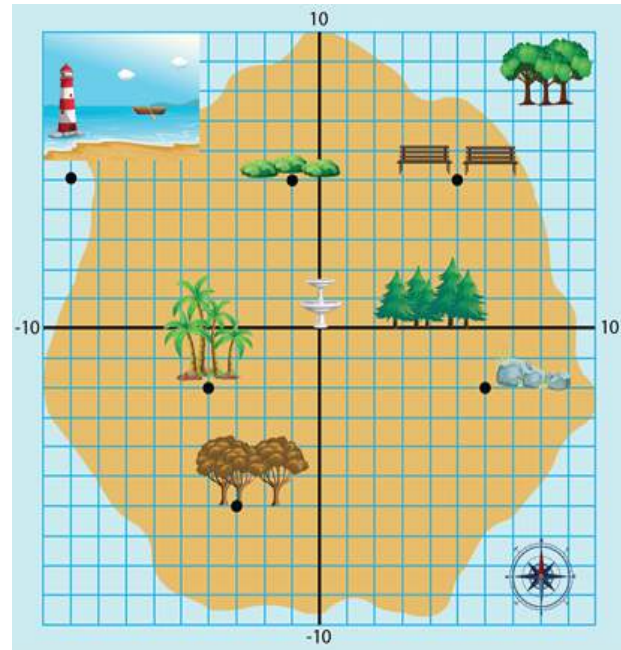


Fig. 4.2, Treasure map

Scale: 1 unit = 100 m on both the axes, with the fountain at the centre being taken as the origin

Question 4

What is the location of the coconut trees?

a. $(-4, -2)$

b. $(-4, 2)$

c. $(4, -2)$

d. $(4, 2)$

Answer

Question 5

Which of the following is the wrong information, as per the map?

- a. Stones are at a distance of 600 m from the vertical axis.
- b. Stones and coconut trees are at the same distance from the x-axis.
- c. We have to walk an equal distance along both the axes to reach the benches.
- d. Pine trees have their x co-ordinate as zero.

Answer

Question 6

The key is buried at a point two-fifths of the distance from the point marked near the bench to the coconut trees. Then the co-ordinates of the key are_____.

a. (-1.4, 2.2)

b. (1.4, 2.2)

c. (1/5, 1)

d. (7/5, 19/5)

Answer

Question 7

The pirate is sitting in his boat, which is 6 units to the west and 7 units to the north from the fountain and is also chasing the key. How far is he from the key?

Answer

Case Study C - Designing a Board Game

Arjun's sister, Ritika, is learning about collinear points. Arjun designs a game, using a handmade chess board. In this game, Player 1 puts two pins at any two points on the chess board, where the lines intersect. Player 2 has to put the third pin, so that all the pins make a line.

To demonstrate this, Arjun placed two pins on the board and asked Ritika to place the third pin by bringing her eye to the level of the pins. A little while later, Arjun's friend, Ali, shows up. So Arjun challenged Ali to place the third pin without bringing his eye to the level of the pins.

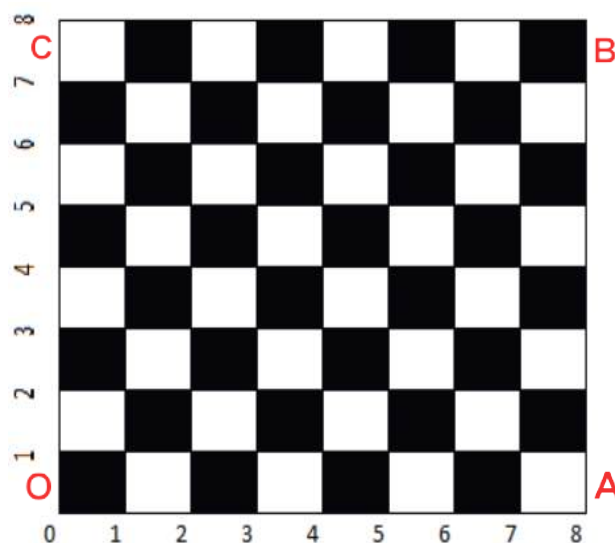


Fig. 4.3, Hand made chess board

Question 8

Arjun puts his pins (P1 and P2) at the following locations: 1 space right and 3 spaces forward from the bottom/near-left corner (the Origin), 3 spaces right and 4 spaces forward from the same corner. Ritika puts her pin (P3) 1 space left and 1 space backward from the top/far-right corner. Are the 3 pins collinear? Justify your answer with mathematical reasoning.

- a. Yes, they are collinear
- b. No, they are not collinear

Answer

Question 9

Ritika in turn challenges Arjun to place his pin (P4) on the bottom reference line (OA), such that P2 is at the same distance from P3 and P4. Where should Arjun place his pin, P4, to satisfy this condition? [Remember that the previously placed pins are not disturbed]

a. at the origin

b. (6, 0)

c. (0, 6)

d. either a or b

Answer

Question 10

Ritika challenges Ali to place a pin (P5) so that the top right corner is the midpoint of the distance between pin P3 and P5. What will be the location of pin P5? Fig.4.4 shows the same chess board with extended axes on all four sides by 2 units. The position of P5 may also be one of the newly formed points.

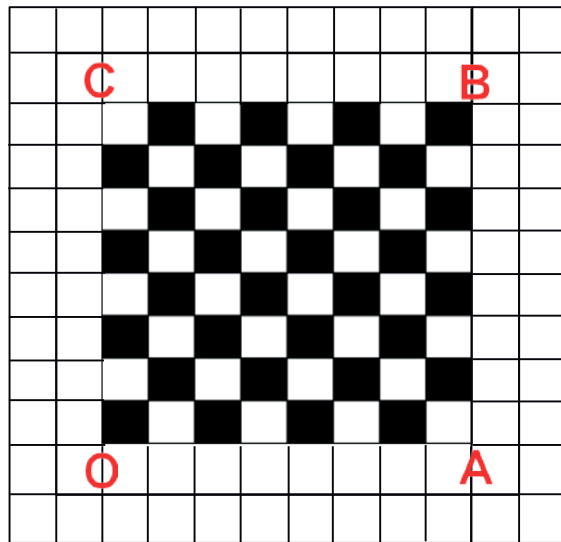


Fig. 4.4, Chessboard with extended axes

a. (9, 0)	b. (9, 9)	Answer
c. (1, 1)	d. (8, 8)	

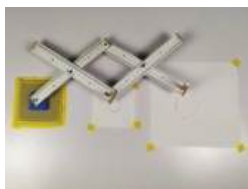
Question 11

Ritika noted the location of pins P1, P2, P3, P5 and calculated the distance between these pins. Which of the statements below is correct?

- a. P1 to P3 distance = P2 to P5 distance
- b. P1 to P3 distance > P2 to P5 distance
- c. P1 to P3 distance < P2 to P5 distance
- d. P1 to P3 distance = P2 to P5 distance = 6 units

Answer

Exploration Pathway



Pantograph

Using a few cardboard pieces, screws and ruler templates, make your own pantograph, which can enlarge/shrink a shape you draw. If you treat the pivot of the pantograph as the origin, predict, observe and validate the position of the points on the bigger and the smaller shapes.



Co-ordinates -
Chess Board

Co-ordinate geometry, also known as analytic geometry or Cartesian geometry, is a field in mathematics where geometry is studied using a co-ordinate system. It has wide applications in the real world: in physics, engineering, aviation, spaceflight etc. It is a simple system, whereby you have co-ordinate axes, and are able to define any single point in space in terms of these co-ordinates. Here, we use a graph sheet to replicate a chess board to validate the distance and section formulas

Theme 5: Introduction to Trigonometry



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Trigonometric Ratios
- Determining the unknown sides of a Right Angled Triangle when one side and an Angle is given
- Area of a Rectangle
- Perimeter of a Triangle
- Tangents from an External Point to a Circle
- Volume of a Cylinder

Case Study A - Raising a Flagpole

A simple way of raising a small flagpole for temporary purposes is by using three guylines around the pole, without burying the foot of the pole deep inside the ground. A guyline is a piece of strong rope tied taut between the flagpole and three stakes, which are equidistant from the pole. One end of the guyline will be tied to the pole at a height of about $\frac{3}{4}$ th of the total height of the pole and the other end to the stake.

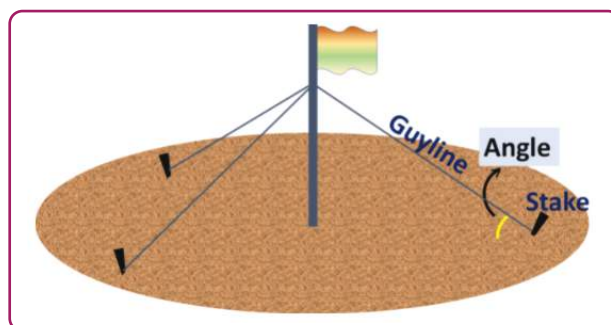


Fig. 5.1, Flagpole using guylines

Sahil wanted to use the 16 ft long pipe, which he had, as the flagpole and wanted to buy rope for making the 3 guylines. As he didn't know the distance from the pole at which he should fix the stake, he tried to figure it out by searching on the internet. He came to know that in order to ensure the stability of the pole, the angle between the ground and guyline

(angle in Fig. 5.1, say θ) should be 15 degrees and the maximum angle one can go up to is 25 degrees.

θ	Sin θ	Cos θ	Tan θ
15°	0.26	0.97	0.27
25°	0.42	0.90	0.47

Table 5.1, Trigonometric values

Question 1

Between 15° and 25°, in which case can Sahil get his work done by buying the minimum length of rope?

- a. 15°
- b. 25°
- c. Length of rope does not depend on the angle

Answer

Question 2

What is the minimum possible length of each guyline, such that the angle between the guyline and the ground still falls in the safe range for the maximum stability? Refer to the safe range of the angle in the background information, which is based on Sahil's research.

Answer

Question 3

If the length of the guyline is represented by L and the angle made by the guyline with the ground by A , what would be the distance between the pole and the stakes?

- a. $L \times \sin(A)$
- b. $\frac{12}{\tan(A)}$
- c. $12 \times \tan(A)$
- d. $\frac{L}{\cos(A)}$

Answer

Question 4

If the pipe used as the pole was 20 ft long, what would be the total length of rope Sahil would need to buy? Assume that he chooses the angle at which he needs the minimum length of the rope and also consider 10% extra rope for tying knots.

Answer

Case Study B - Styles of Roofs

The steepness of a roof is called the roof pitch. Whenever the roof is not flat, in construction terminology, the roof is said to be pitched. Carpenters will frame the rafter of the roof at an angle with the horizontal to “pitch” a roof. As the pitch of the roof increases, there will be a corresponding increase in the materials needed for framing and sheathing. Among the different types of roofs, gable roofs are the simplest form of roofs. The diagram below shows a simple structure with a gable roof.

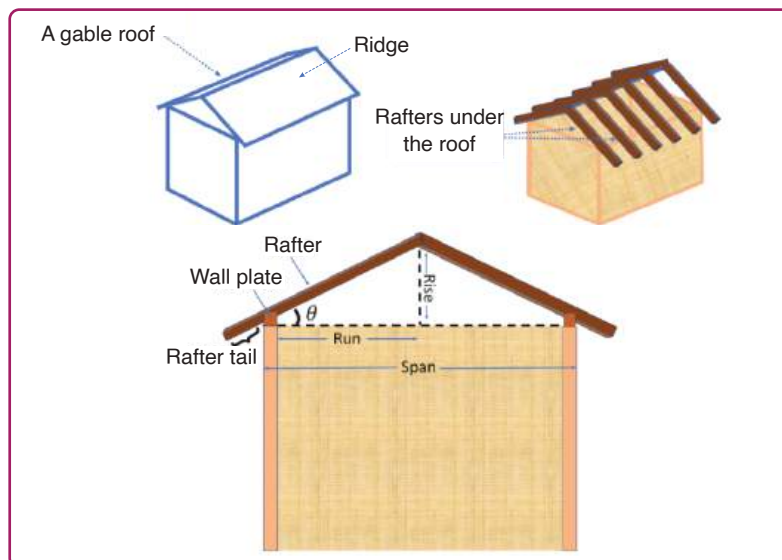


Fig. 5.2, Gable roof, the simplest form of roofs

The line at which the two roof planes meet is called the ridge. The horizontal distance between the two walls is called the span. The horizontal distance between the wall and the point under the ridge is called the run, which is also equal to half the span. The vertical distance between the ridge and the line joining the top of the two walls is called the rise. The rafter is the wooden beam or the metal frame, which runs from the ridge to the wall plate on the wall. The exterior part of the rafter, which extends beyond the wall, is called the rafter tail.

Question 5

Speaking in terms of numbers, the pitch represents how much a roof rises as it moves in from the wall. So typically, the pitch is represented by rise over run, i.e. if the rise is 3 units and the run is 4 units, then the pitch is $\frac{3}{4}$. This is also called the slope of the roof. If θ is the angle between the rafter and the line joining the walls, then which of the following correctly represents the pitch/slope?

a. Cot θ	b. Tan θ	Answer
c. Cos θ	d. Sin θ	

Question 6

Thatching is a technique to build a roof with dry plant material, such as reeds and straws (dry stalks of cereal plants, water plants etc), palm branches and leaves, etc. A thatched roof needs to be very steep in order to drain properly. We can also see stone roofs in some old structures. Stone roofs can't be very steep, so as to avoid the stones from falling down.



Fig. 5.3, A thatched roof, stone roof and a metalsheet roof.

Image of thatched roof by Eric Christensen (own work) via Wikimedia Commons, stone roof by ЈованВБ (own work) via Wikimedia Commons, metal sheet roof via flickr.com

Fig. 5.4 shows three schematic diagrams representing a thatched roof, stone roof and a metal sheet roof.

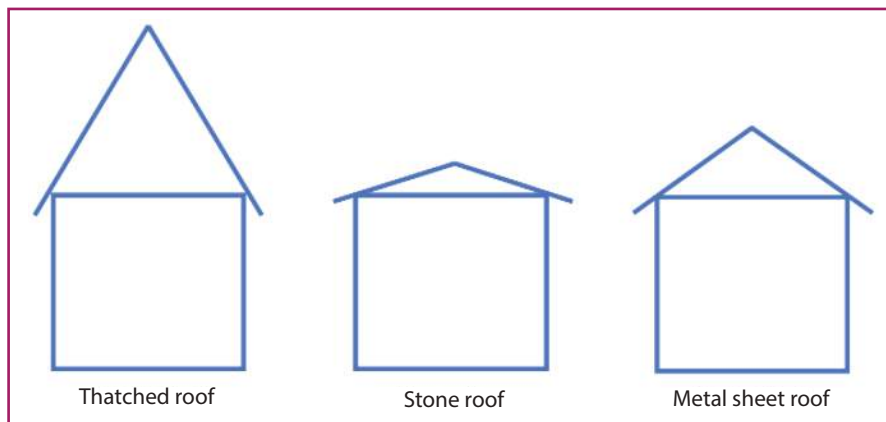


Fig. 5.4, Schematics representing the gable roof made of different materials

Which of the following correctly represents the decreasing order of the value of θ , if the span of all three structures in Fig. 5.4 are the same?

- a. Stone roof > Thatched roof > Metal sheet roof
- b. Thatched roof > Metal sheet roof > Stone roof
- c. Stone roof > Metal sheet roof > Thatched roof
- d. Metal sheet roof > Stone roof > Thatched roof

Answer

Question 7

Sahil wanted to make a gable roof for a building in his farm. He thought of using tiles to make the roof. The span of his building is 15.2 feet. In his region, it rains heavily. He decides to have the angle θ , between the rafter with the horizontal, be 40° .

- i. What should be the length of the wooden block to make the rafter, assuming a rafter tail of 2 feet? Give your answer to the nearest whole number.

Answer

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
40°	0.64	0.76	0.83

Table 5.2, Trigonometric values

- ii. When Sahil goes to buy the tiles he finds out that the cost of metal sheets is less as compared to tiles. But the disadvantage of having a metal roof is that, there will be a rise in the interior temperature during summer, as metal is a good conductor of heat. So he wanted to experiment by adding a layer of plywood underneath the metal sheet covering. The room which these roof covers has a base of 14 feet x 15.2 feet.

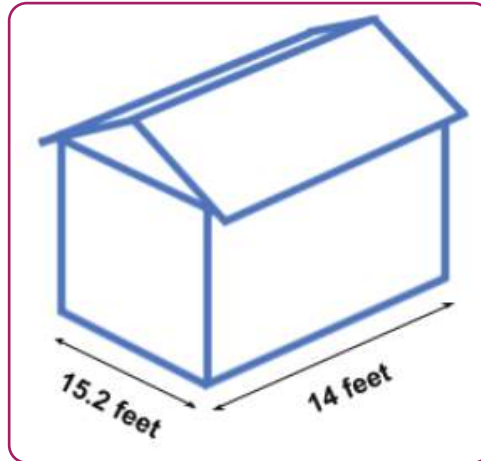


Fig. 5.5, Diagram showing dimensions of the building on Shyam's farm (drawing not to scale)

The region where Sahil lives, plywood is available in the form of sheets of the following standard sizes:

8 feet x 3 feet, 8 feet x 4 feet,
7 feet x 3 feet,
6 feet x 4 feet, 6 feet x 3 feet

Sahil wants to estimate the cost before ordering the raw material. But before that, he wants to calculate the minimum number of plywood sheets that he should buy. What is the minimum number of plywood sheets in total that Sahil will need to make the two sides of the roof, and which sizes should he select to ensure there is no wastage? Assume the false roof made of plywood is also a gable roof and its area is equal to that of the main roof made of metal sheets.

Answer

Case Study C - Optimal Packing

Ria and Avni are designing a triangular storage box for their tennis balls, as shown in Fig. 5.6. The diameter of each ball is 6.8 cm. Ria thought that to calculate the length of the inner side of the box, she could just multiply the diameter of the balls by 5. But Avni pointed out that it is not possible because there is an extended part beyond the last ball on both the sides. Both of them racked their brains and remembered what they have studied about tangents to a circle.

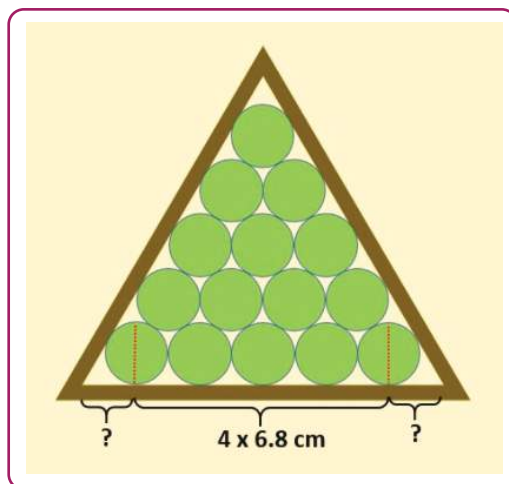


Fig. 5.6, Tennis balls inside a triangular box

They recalled that whenever two tangents are drawn from an external point to a circle, the two triangles formed by the tangents, the radii and the central line together are congruent to each other. In fig. 5.7, $\triangle POA \cong \triangle POB$. Hence $\angle APO = \angle BPO$. In other words, the line PO bisects $\angle APB$.

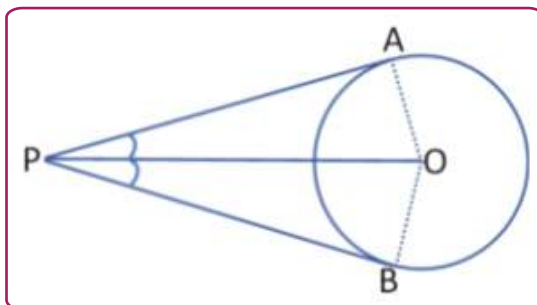


Fig. 5.7, Tangents drawn to a circle from an external point P.

Question 8

Which of the following expressions will give the exact length of the inner side of the triangle?

a. $(6.8 \times 4) \text{ cm} + (3.4 \times \sin 30^\circ) \text{ cm}$

b. $(6.8 \times 4) \text{ cm} + (6.8 \times \cos 30^\circ) \text{ cm}$

c. $(6.8 \times 4) \text{ cm} + (3.4 \times \tan 30^\circ) \text{ cm}$

d. $(6.8 \times 4) \text{ cm} + (6.8 \times \cot 30^\circ) \text{ cm}$

Answer

Question 9

If Ria decided to make another triangular box, this time to fill 15 balls of diameter 5.7 cm, the inner perimeter of the box will be close to _____.

θ	$\sin \theta$	$\cos \theta$
30°	0.5	0.86

a. 45 cm

b. 75 cm

c. 84 cm

d. 98 cm

Answer

Sextant

A sextant is an age-old device, used by astronomers, seafarers and engineers for millennia, to measure the angle of elevation of different objects as seen by you. You can make your own simplified version of a Sextant (often called the Clinometer), using a plastic straw, thread, some weight and a protractor template.

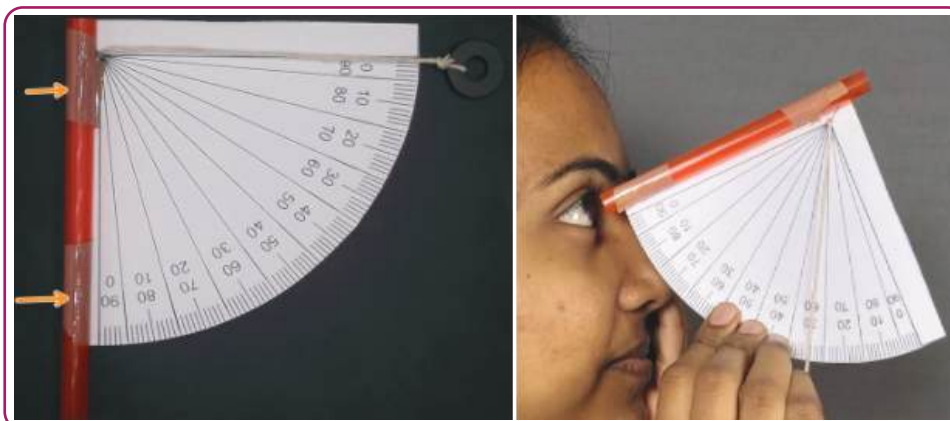


Fig. 5.8, Handmade sextant - a simple device to measure angle of elevation

Case Study D - Estimation

Shyam is standing near a huge overhead water tank at a distance of 35 m from it. He had his own handmade sextant with him. With his sextant he noticed that the angle of elevation of the top and bottom of the tank from his eye level is 30° and 23° , respectively. Based on the distance between opposite pillars under the tank, Shyam estimated the inner diameter of the tank to be 10 metres.

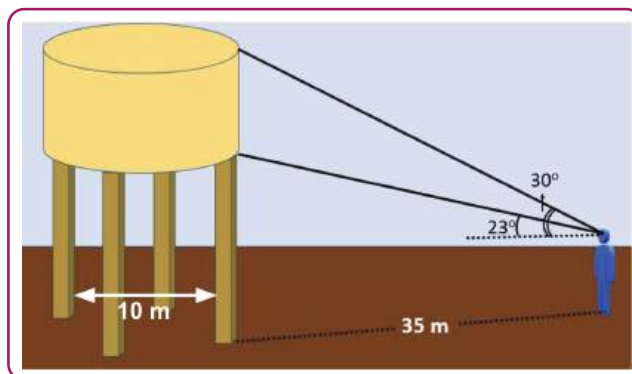


Fig. 5.9, Angle of elevation of the top and bottom of the overhead water tank

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
23°	0.39	0.92	0.42
30°	0.5	0.86	0.57

Table 5.4, Trigonometric values

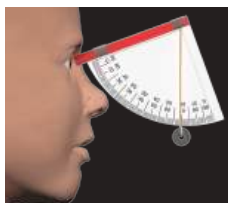
Question 10

The total amount of water that can be contained in that tank is close to _____

Note: $1\text{m}^3 = 1000$ litres

a. 21000 litres	b. 232000 litres	Answer
c. 412000 litres	d. 520000 litres	

Exploration Pathway



Sextant Model

A sextant is an age-old device, used by astronomers, seafarers and engineers for millennia, to measure the angle of elevation to different objects as seen by you. Here, you make your own Sextant using a plastic straw, thread, some weight and a protractor template.

Theme 6: Surface Areas and Volumes



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Surface Areas and Volume Formulae for Basic Solids - Cube, Cuboid, Cylinder, Cone, Frustum of Cone, Sphere, and Hemisphere
- Surface Area and Volume of an Object formed by combining two or more Basic Solids.

Case Study A - Greenhouse Design

Have you ever wondered why a car kept in the Sun is super-hot inside? Why glass buildings waste a lot of money on electricity bills due to air conditioning? Well, it's because glass traps heat and makes the interiors even hotter than outside. Farmers have used this technique - especially in cooler climates - to grow, say, tropical plants in temperate regions. These transparent structures where these plants are grown, indoors, are called greenhouses. Now you also might make the link why we call the warming of the Earth due to certain gases in the atmosphere as the 'greenhouse effect', or why the culprit gases are called 'greenhouse gases'!

Anyway, back to greenhouses themselves: since we can control the temperature within them and keep them especially warm, it allows botanists and farmers to grow crops or ornamental plants year-round, and keep them safe from a lot of the natural factors outdoors. The greenhouse structure is typically constructed using aluminium or wood, and while glass provides the best 'greenhouse effect', it also weighs and costs a lot, so plastic substitutes are also used for the transparent covering.

So the next time you go to a nursery or a botanic garden, and see a greenhouse, go in and enjoy the micro-climate experienced inside and also notice the kind of structure and materials used to make them.

The questions below are related to the design of greenhouses.

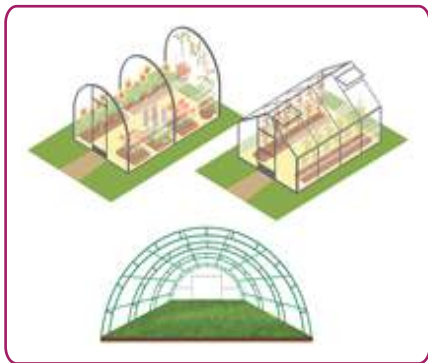


Fig. 6.1, Sample designs of greenhouse

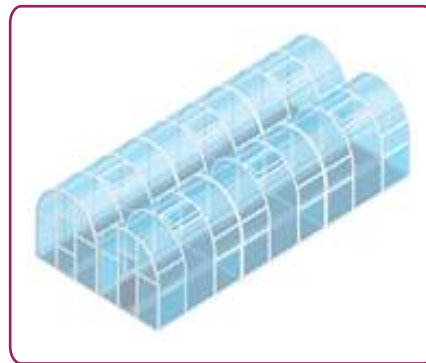


Fig. 6.2, Two or more greenhouses are sometimes joined side by side so that they have fewer external walls, and the material costs are consequently less. A greenhouse has a wide area of glazing on its sides and roof so that the plants are exposed to natural light for much of the day.

Question 1

A farmer constructed a greenhouse with the dimensions, as shown in Fig. 6.3. He needs to buy transparent tarpaulin to cover it. Transparent tarpaulin comes in 2 m x 20 m rolls and most vendors do not provide partial rolls. How many rolls does the farmer need to buy? Note that he keeps the entire front portion open to access the greenhouse. Except the front portion, he wants to cover the entire structure with tarpaulin. He assumes that there won't be any wastage when he fixes the transparent tarpaulin.

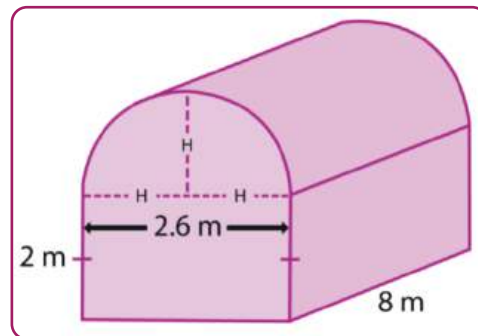


Fig. 6.3, Front view of the design of a greenhouse. Note that $H = 1.3$ m here.

a. 1 roll	b. 2 rolls	Answer
c. 3 rolls	d. 4 rolls	

Question 2

As the farmer started fixing the transparent tarpaulin on the greenhouse, he realised that he had to cut the rolls into small pieces to cover some parts of the greenhouse, which helped him minimise wastage. This was also necessary, because some parts required both lengths of the sheet to be more than 2 m, whereas the transparent tarpaulin rolls had a maximum length of 2 m on one side. He labelled different parts of the greenhouse as shown in Fig. 6.4 & 6.5

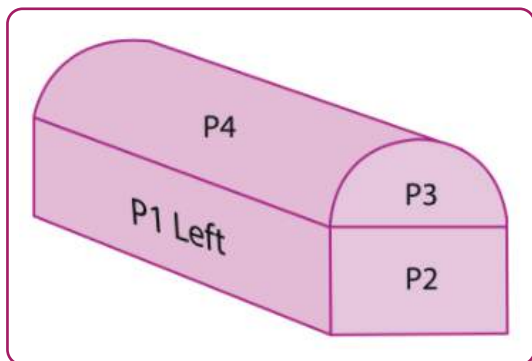


Fig. 6.4, Parts of the greenhouse labelled

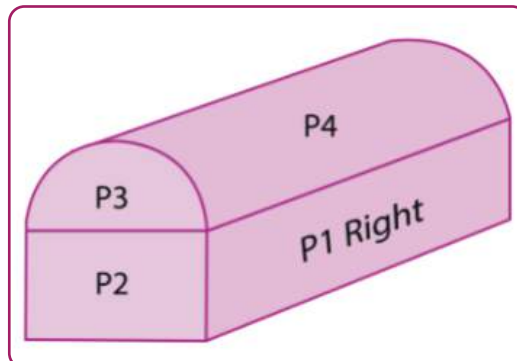


Fig. 6.5, P1 Right labelled separately

Note that he labelled the left and right side of P1 as two separate parts, so that separate sheets are cut out for both parts.

He then decided to cut rectangle-shaped pieces for each of these parts. The dimensions of the pieces P1 Left, P1 Right and P2 have been filled in boxes next to the respective sides of the rectangles. In the same manner, fill the dimensions of the remaining pieces.

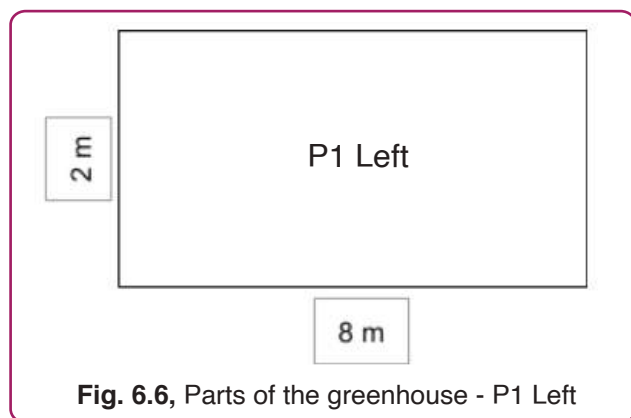


Fig. 6.6, Parts of the greenhouse - P1 Left

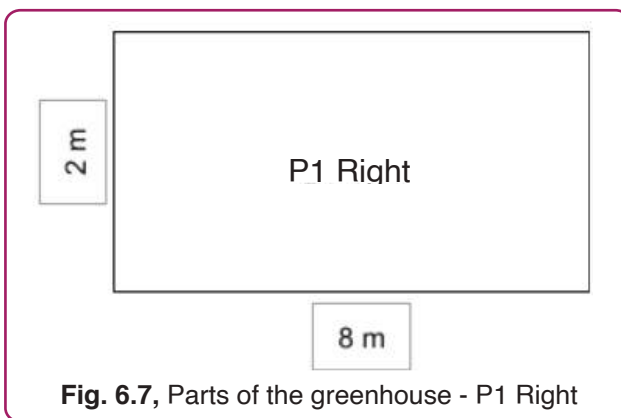


Fig. 6.7, Parts of the greenhouse - P1 Right

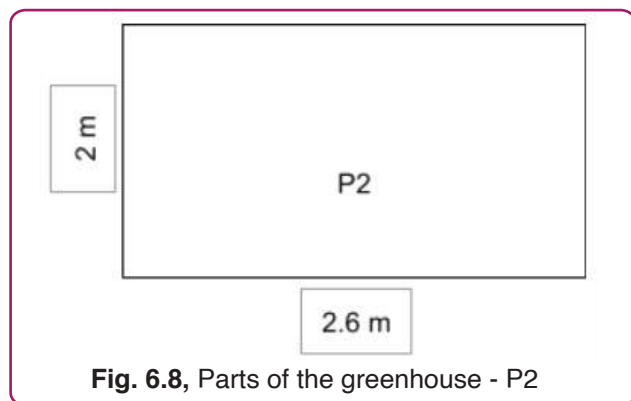


Fig. 6.8, Parts of the greenhouse - P2

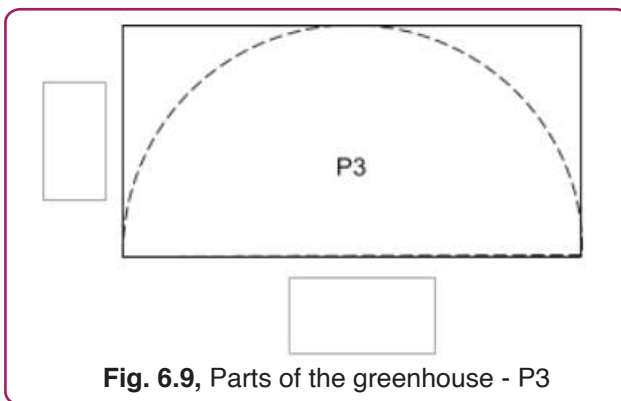
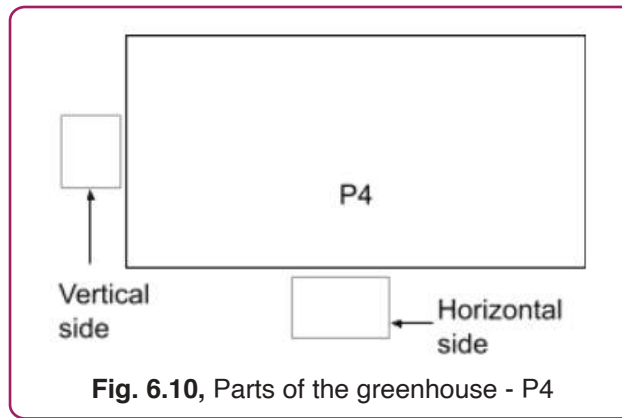


Fig. 6.9, Parts of the greenhouse - P3

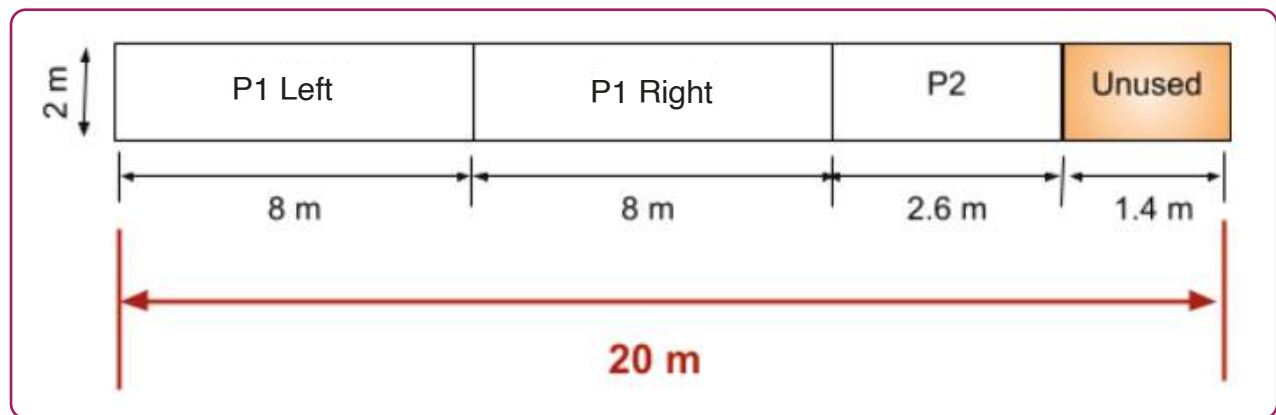


- i. P3 vertical side: _____ iii. P4 vertical side: _____
- ii. P3 horizontal side: _____ iv. P4 horizontal side: _____

Question 3

As he details the plan further, he chose to leave a 5 cm gap on both sides, between the roof and the wall, for ventilation. He realises that this reduces the required length of the P4 vertical side by 10 cm.

He now started laying out these 5 parts on the rolls. An example layout has been shown in Fig. 6.11. Note that this is only an example.



Some parts had to be cut into smaller sub-parts so as to fit into the dimensions of the roll and to minimise wastage.

- i. Going by his method of cutting rectangular pieces, what would be the minimum number of rolls he would need?

Answer

ii. What is the area of the transparent tarpaulin sheet which would go to waste?

Answer

Question 4

Apart from controlling the temperature, and the watering of the plants, greenhouses also need to be well-ventilated. Plants, like us, need a constant supply of fresh air, and having a good ventilation system allows you to control temperature and humidity better and reduce the proliferation of pests. Effective ventilation can be achieved naturally by having more vents and windows, or via a powered system that circulates air. In a powered system, there has to be a combination of fans and vents so that the outside air is able to replace the inside air within minutes. The power and number of these fans is determined by the size of the greenhouse.

Calculate the maximum volume of air in this empty greenhouse structure, shown in Fig. 6.3?

Answer

Question 5

A hollow metal pipe with an external diameter of 28 cm is 1 cm thick and is placed at the centre of this greenhouse along its longest length. The farmer plans to use it as a partitioning line by shifting it in between two varieties of plants. How many people will be required to move the pipe? Assume one adult can lift a maximum of 50 kg. [Consider the density of this metal as 8.05 g/cm^3]



Fig. 6.12, Metal pipe used as separator in the greenhouse

a. 5 or less than 5	b. 6 to 9	Answer
c. 11 to 12	d. More than 12	

Case Study B - The Peanut Vendor

India has a large number of small entrepreneurs who choose to run their own small businesses instead of taking up a job in a larger organisation. A typical Indian market would have multiple small streets with a large number of small shops on each street. Many of these are mobile shops, with the vendor moving her/his products on small carts and reaching out to the customers. One such unique and often sought after vendor is the peanut vendor, who carries the peanuts as well as the equipment required to roast the peanuts on a cart. Some of these carts are two- or three-wheeled carts. Other than the hot and crispy roasted peanuts, these vendors are a source of attraction, especially for the younger crowd, because of the perfect cones they make from waste papers to serve the peanuts to their customers. No glue is used while making these cones. Why do they make cones and not any other shape, say a cylinder or cuboid?

Question 6

Lakshmi and Ira buy some roasted peanuts from a street vendor who makes cones from papers, which have an almost circular shape. As shown in Fig. 6.13, around $1/4^{\text{th}}$ of the circle is used to overlap with another $1/4^{\text{th}}$ part of the circle. So the surface area of the cone would be $3/4^{\text{th}}$ of the circle.

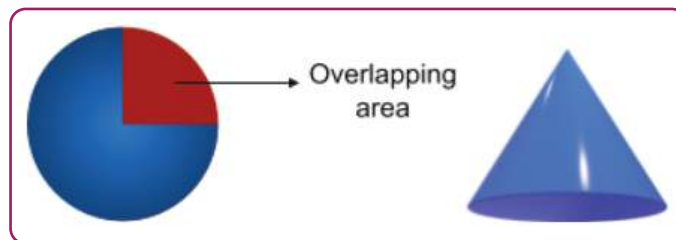


Fig. 6.13, Circular paper used to make a cone

Lakshmi and Ira insisted on being served the peanuts in two separate cones. The vendor cut the circle into two semicircles and taping the edges together he made two cones.

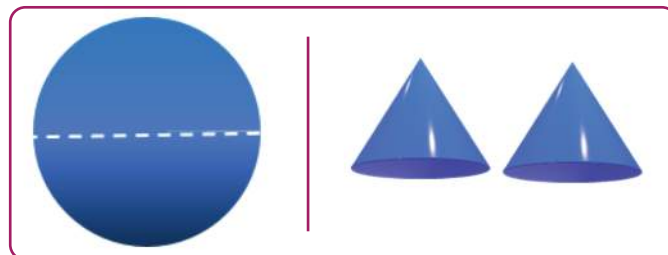


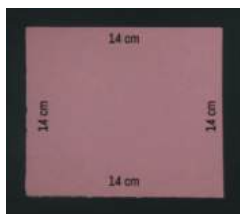
Fig. 6.14, Circle cut into half to make two smaller cones

Do you think he gave more peanuts when compared to a single cone or less or equal? Assume that the peanuts are packed as tightly as possible in both the cases.

a. More	b. Less	Answer
c. Same	d. Depends on the size of the circle	

Case Study C - Activity to Visualise Volume

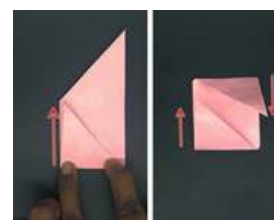
So as to visualise volume, one can make a regular cube. Given below is one of the ways of doing this, using 6 square papers.



Step 1 - Take a square sheet of paper of size 14 cm x 14 cm.



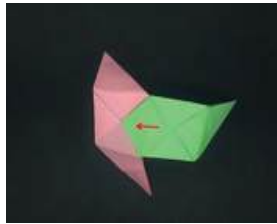
Step 2 - Fold the sheet to make one parallelogram shaped module



Step 3 - Fold triangle shaped ends to create flaps of the module.



Step 4 - Repeat the steps with 6 square sheets to create 6 modules.



Step 5 - insert one module's flap into the pocket of another, as shown.



Step 6 - Link other 4 modules in a similar manner to make a 6 sided cube.

Question 7

The volume of the cube hence made was measured using the process shown in Fig. 6.15 and Fig. 6.16.



Fig. 6.15, Cube opened and a plastic cover arranged inside



Fig. 6.16, Fill the cube upto the brim and measure the volume of collected water using a volume measuring beaker

Following observations were made and recorded, using the cube thus made.

Volume of the cube	125 ml
Length of the side of the cube	5 cm
Length of each of the square papers from which we made the faces of the cube	14 cm

- i. If we want to make a cube-shaped box, which has a volume of 1 litre, using the above process, what should be the length of the side of the box when compared to the length of the box made above?

a. 8 times more	b. double	Answer
c. 4 times more	d. half	

- ii. What should be the length of the 6 square papers for making the 1 litre cube? Note that the length of each square paper used to make the above cube was 14 cm. The process of making one module from a square sheet has been shown in Fig. 6.17.

a. 112 cm	b. 28 cm	Answer
c. 56 cm	d. 7 cm	

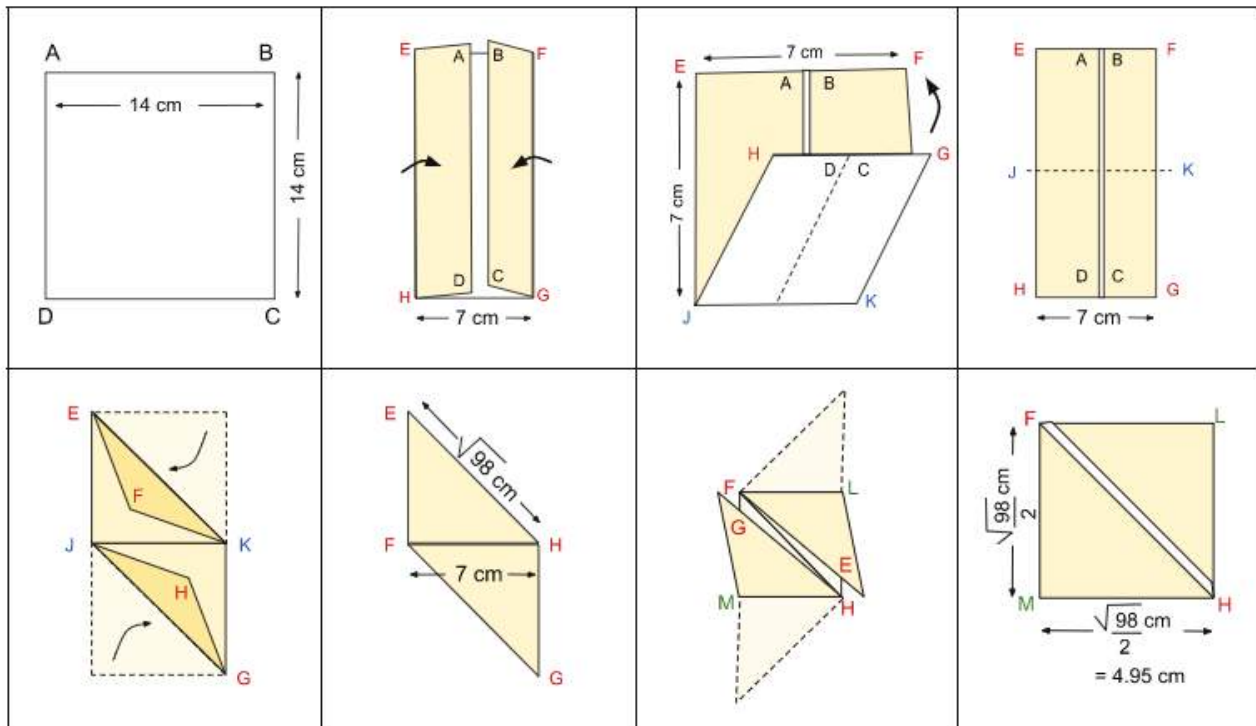


Fig. 6.17, Square sheet of 14 cm x 14 cm folded to make one module with square in the middle and triangular flaps. The side of the square in the middle would be 4.95 (can be rounded off to 5) cm long.

Case Study D - The Selling Trick

Way back in the 1950s, a toothpaste manufacturing company is said to have conducted a competition among their employees to come up with ideas to improve their sales. After looking at multiple ideas, they received a note from an outsider who offered to sell his idea for \$100,000. At that time this was a huge sum of money, so the company ignored it initially. However, the stranger showed up at the office and confidently claimed that his idea was infallible. The company was convinced and decided to buy the idea. Next day the company received a brown envelope from this gentleman, which contained a small slip of paper saying, “Make the opening bigger”.

Question 8

Consider a toothpaste tube, which contains around 50 ml of toothpaste. The inner diameter of the tube's opening is 5 mm and the company decides to increase the diameter by 1 mm. By what percentage would the amount of toothpaste coming out in one usage increase? Assume that the length of toothpaste squeezed out in each usage remains the same as before. Write your answer in the box below.

Answer

Question 9

Most consumers don't pay attention to the volume of toothpaste squeezed out in each usage. They tend to just fill the entire length of the toothbrush. This could be the effect of habit or the visual cues from advertisements showing the entire length of the toothbrush being filled with toothpaste.



Fig. 6.49, Toothbrush with toothpaste

In a family of 3 members, how many days will a 50 ml toothpaste last? Consider the length of the toothbrush to be 2 cm for all the three family members. Also assume they brush twice in a day and fill the entire length of the brush during each usage. Consider the inner diameter of the toothpaste tube to be 5 mm. Write your answer in the box below. Note that $1 \text{ cm}^3 = 1 \text{ ml}$.

Answer

Question 10

The toothpaste company tied up with a popular toothbrush company and increased the length of the toothbrush by 25%. Now, with the increase in the toothpaste tube's inner diameter to 6 mm and increase in the length of the toothbrush, how many days will a toothpaste tube of the same volume last for the 3-member family? Assume that all the family members replace their toothbrushes, thanks to the free toothbrush offer on the new toothpaste. Write your answer in the box below.

Answer

Exploration Pathway



Volume and
Surface Area

Take two cardboard pieces of rectangular shape with the same dimensions and one side being longer than the other. Roll one rectangle along the shorter side and another along the longer side and make two cylinders. Predict and validate which cylinder has a larger volume and which one has more surface area. Pour soil in both cylinders to compare which one has more volume. Repeat this exercise by cutting one rectangle into two equal parts and comparing the total volume of two cylinders made from the half rectangles with that made from the full rectangle.



Visualise Volume

As our brains go from grasping one dimension to two to three, certain aspects of our understanding become less and less clear. However, volume is one of our most routine day-to-day units, but causes innumerable conceptual misunderstandings. In this incredibly visual TACTivity, you make your own cubes of fixed volume, using coloured paper to gain a better grasp of the concept of volume.

Theme 7: Statistics



Prior Knowledge

It is recommended that you revise the following topics before you start working on these questions.

- Mean, Median and Mode for grouped frequency data.
- Representation and interpretation of the cumulative frequency distribution graph.
- Finding the median of grouped data graphically from the less than type and the more than type ogive.



Statistics - A Handy Tool

Statistics pervade almost every facet of our life: from collecting data about population growth, to looking at Covid-19 numbers, to analysing Virat Kohli's "average", we use numbers to justify a whole host of opinions and crucially, also use them to make all sorts of decisions. Statistics, and its close cousin Probability, are useful in research, economic planning, collecting and interpreting public data etc. Statistics is also one of those subjects where 'bigger' or more is better! The more data you collect, the more accurate your analysis will be. Statistics can sometimes be misleading. For example, Virat Kohli's current average in Test cricket may be 50 runs per innings, however, when he goes out to bat, he is most likely to get out for a score under 20. He crosses 50 only once in 3 innings. So which statistic you use for what purpose, and to make what point, also becomes crucial.

Case Study A - Chillies per Plant

A study of the yield of 200 chilli plants is recorded by a farmer and is given in Table 7.1.



Fig. 7.1, Chilli Plant

Chillies per plant	1-5	6-10	11-15	16-20	21-25
Number of plants	15	14	75	87	9

Table 7.1, The number of plants for chillies per plant

To avoid excessive usage of chemicals in his farm, he wants to buy organic manure, only if the overall number of chillies per plant is less than 12. Analyse this data and answer the question given below.

Question 1

- i. Estimate the mean and median for the given data without actual calculations?

Answer

- ii. To help the farmer make his decision, which central tendency (mean, median or mode) will you find out. Calculate its actual value.

Answer

Question 2

How many plants are there that yield less than 21 chillies per plant?

a. 87	b. $200 - 9$	Answer
c. $15+14+75$	d. 9	

Question 3

He uses organic manure only on those chilli plants which are yielding less than 5 chillies now, and with this the new modal class changes to 11-15. Which of the following statements cannot be true? (There may be more than one correct answer)

- a. All the chilli plants that were giving 1-5 chillies now give 11-15 chillies
- b. At least 13 chilli plants that were giving 1-5 chillies now give 11-15 chillies
- c. Half the chilli plants that were giving 1-5 chillies now give 11-15 chillies
- d. All the chilli plants that were giving 1-5 chillies now give 21-25 chillies

Answer

Question 4

If the farmer decides to plant more chilli plants, rather than using organic manure, to increase the mean value of crop production, then

- a. Mean will further reduce if those new plants yield less than 10 chillies
- b. Mean will increase just by planting more chilli plants, irrespective of chillies per plant
- c. Mean is not dependent on the number of plants; it depends only on the total number.
- d. Mean is not dependent on the total number of chillies; it just depends on the number of plants.

Answer

Case Study B - Cycling Race

In a marathon cycling race, the organisers had arranged resting stalls at regular intervals where the cyclists can stop for a quick bite, water, glucose, etc. Each resting stall had a person appointed to track the number of cyclists crossing that point and their speed. He had to relay back these informations to the organisers. One such resting stall, just after the toughest section of the cycling path, which is near the halfway point, was used to analyse this speed data collected. The core management team of the organisers were stationed at the starting point and they continued to plot the cumulative frequency graph of the speed of the cyclists, using a software. The team continued to analyse this graph during the race because it had an impact on the day's schedule.

Question 5

To conduct the event smoothly and in a timely manner, the management team wanted to know: (1) When will the first 3 cyclists reach the venue, so that they can start the award ceremony and (2) The time at which the last cyclist reaches, so that they can close the event.

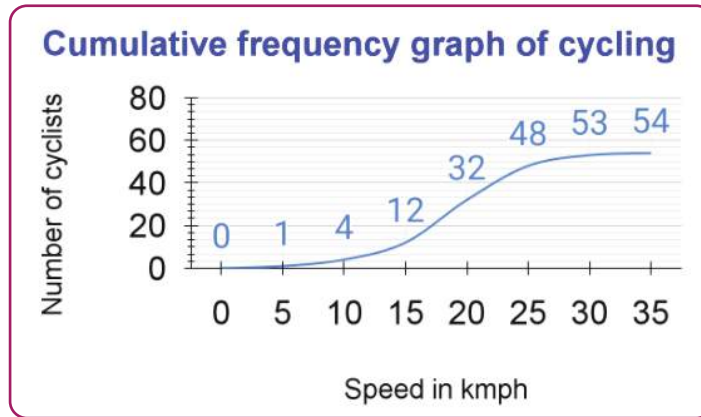


Fig. 7.2, Cumulative frequency graph of cycling speed

- i. Based on the current cumulative graph (Fig. 7.2), what is the speed (in kmph) of the cyclists who will help them answer the first question? In the options given below, assume that both numbers of the range are included.

a. 48 to 54	b. 30 to 35	Answer
c. 25 to 35	d. 52 to 54	

- ii. What is the speed (in kmph) of the cyclist who will help them answer the second question?

a. $0 < \text{speed} < 1$	b. $0 < \text{speed} < 5$	Answer
c. 5	d. $0 < \text{speed} < 10$	

Question 6

Find the number of cyclists who are travelling faster than 25 kmph.

a. 10	b. 48	Answer
c. 6	d. 16	

Question 7

Table 7.2 lists the speed of cyclists and the number of cyclists. Find out how many cyclists are there with speeds 10-15 kmph and 20-25 kmph?

Speed of cyclist in kmph	Number of cyclist
0-5	1
5-10	3
10-15	x
15-20	20
20-25	y
25-30	5
30-35	1

Table 7.2, Speed of cyclist and the number of cyclists

Answer

Case Study C - Flight Delay

On a particular day, due to bad weather conditions, all the flights were delayed in an airport and the maximum delay time was nearly an hour. The details about the flight delay times were analysed by referring to the frequency of delayed flights as given in Table 7.3. It was noted that a total of 45 flights were delayed in the period of 6 hours after which the weather conditions came back to normal.

Time delayed (minutes)	Frequency
0-10	8
10-20	10
20-30	17
30-40	3
40-50	6
50-60	1

Table 7.3, Time delayed and frequency

Question 8

What is the average delay time, in minutes?

a. 23.22	b. 30	Answer
c. 19.22	d. 8	

Question 9

How many flights were delayed beyond half an hour?

a. 10	b. 7	Answer
c. 27	d. 8	

Question 10

What is the delay time, in minutes, which has an equal number of flights that are delayed below and above this time?

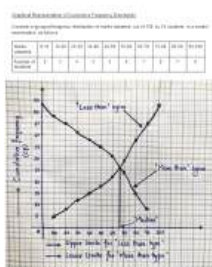
a. 17.36	b. 22.64	Answer
c. 30	d. 25	

Question 11

What is the most frequent delay time?

a. 55	b. 33.33	Answer
c. 17	d. 23.33	

Exploration Pathway



Ogive and Median

For the collected dataset, make a less-than type and more-than type cumulative frequency table. Choose a proper scale, based on the data. Plot the lower limit of the class interval along the x-axis and cumulative frequency along the y-axis to get a more-than type ogive. Similarly, on the same graph plot the upper limit of the class interval along the x-axis and cumulative frequency along the y-axis to get a less-than type ogive. Drop a perpendicular from the intersection point of these two graphs to read the median value. Understand the significance of the median value for the given data.



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